Supported kernels, translation lattices and finite marked quotients

Human Compression Project

Abstract

We study marked groups through the intersections of subgroups generated by subsets of their markers. A construction over a group with deletion maps attaches local extensions whose kernels are invisible on proper local subgroups. We prove generation of the full compatible product, compute every parabolic kernel and obtain the intersection property. The same coordinates give local recognition, deletion maps, lower-central identities, separator factorizations and transfer of the flag property for coset geometries. For finite two-groups whose noncommuting diagram is a forest, we identify a canonical class-two quotient and a finite, exhaustive parameter space of marked quotients of any prescribed excess order. Explicit affine square groups supply local models of every required two-power order. The paper includes the supporting quotient, Frattini and amalgam arguments so that subsequent applications can cite complete statements with specified hypotheses.

1 Introduction

A presentation by involutions does not by itself determine a regular polytope or hypertope. The distinguished generators must have the prescribed pair orders, their generated subgroups must intersect correctly, and a geometric conclusion may require a further common-representative condition on cosets. Establishing these properties separately for every presentation can obscure the part of the proof that is independent of its numerical parameters.

The main construction here retains an already understood marked quotient and records the extra relations in separate kernel coordinates. Each attached group is required to agree with the quotient on every proper part of its support. This hypothesis lets us isolate its whole kernel, even when one support contains another. Once the kernels have been isolated, all marked subgroup intersections and orders follow from a single coordinate calculation. The geometric assertion uses an additional proved transfer of typed coset Helly; it is not inferred from the intersection property alone.

A second branch identifies the quotient which every finite two-group marking on a forest must have. The proof uses the intersection property in the original group, a descending lower-central argument and the separation of a forest at an edge. It then reduces classification at any fixed excess order to normal subgroups of one finite nilpotent quotient. The explicit square groups in the affine-square section provide a complementary construction: their orders vary, while their proper marked subgroups remain unchanged.

The companion applications use these mechanisms for prescribed two-power orders, three-generator hypertopes, square-section assemblies, common quotients, high-type classifications and the nondegenerate order-1024 census. Construction and presentation recognition are distinct: the former isolates local kernels in a compatible product, while the latter proves that a proposed presentation has no further kernel. Both steps are proved below. The source results retain their original attribution; no historical priority for these shared arguments is asserted. The same kernel language controls the mixed-prime affine families. An exact Schreier presentation first identifies the infinite translation kernel; lattice periods then determine the quotient and every marked intersection. We give a criterion for arbitrary parabolics, including those with nonzero translation kernels, before specializing to the

faithful finite adjacent subgroups of the rank-three applications. The resulting period-and-local-extension theorem combines this recognition with the supported construction. The complete square normal form and the kernel-isolation proof occur only once here.

This volume retains all seven previously admitted Topic A original contracts and the complete mixed-prime contract L012. It adds no original or conjecture credit. It preserves all source corrections, all eleven whole required program/data files, and the separate retained classical and software inputs. The arbitrary-parabolic lattice criterion is newly stated in this edition; its elementary coset proof and its current lack of full Lean formalization are explicit. All numerical comparisons remain deferred to a separately reviewed whole-volume boundary.

2 Markings, images and kernels

A marked group is a group G together with a finite indexed generating family $(r_i)_{i\in I}$. Except where stated, the markers need not be involutions or distinct. For $S \subseteq I$, put

$$G_S = \langle r_i : i \in S\rangle, \qquad G_\varnothing = 1.$$

We call these subgroups parabolics, retaining their index sets even when two of them happen to coincide. The marking has the intersection property, or IP, if

$$G_S \cap G_T = G_{S\cap T} \quad (S, T \subseteq I). \tag{M1}$$

A marked homomorphism sends each source marker to the corresponding target marker. Its image on G_S is therefore the target parabolic with index set S.

All subgroup products below are products of sets unless a subgroup assertion is made. We write $\gamma_1(G) = G$ and $\gamma_{j+1}(G) = [\gamma_j(G), G]$. We use the elementary correspondence and isomorphism theorems for groups. In the finite p-group arguments we retain Burnside's basis theorem, the nilpotence of finite p-groups, and the usual multilinear commutator description of the lower-central graded Lie ring. Precisely, its degree-one part generates the graded Lie ring, and degree j is generated by left-normed commutators of weight j in any generating set. In the amalgam section we explicitly retain the normal-form theorem for an amalgamated free product. None of these classical inputs is presented as a result originating in this manuscript.

Lemma UA.K0 (lifting one intersection). Let $q : H \to Q$ be a homomorphism and let $C \le A\cap B$, where $A, B, C \le H$. If

$$q(A) \cap q(B) = q(C),$$

then

$$A \cap B = C \quad\Longleftrightarrow\quad A \cap B \cap \ker q \le C. \tag{M2}$$

Proof. The forward implication is immediate. For the converse, let $x \in A \cap B$. Its image belongs to $q(C)$, so choose $c \in C$ with $q(c) = q(x)$. Then $xc^{-1} \in A \cap B \cap \ker q$, which is contained in C by hypothesis. Multiplying by c gives $x \in C$. The inclusion $C \le A \cap B$ gives equality. □

Theorem UA.K (kernel profiles). Let $q : H \twoheadrightarrow Q$ be a marked epimorphism and suppose that the marking of Q has IP. Write $K = \ker q$ and $K_S = K \cap H_S$. Then H has IP if and only if

$$K_S \cap K_T = K_{S\cap T} \quad (S, T \subseteq I). \tag{M3}$$

Proof. Apply UA.K0 to $A = H_S$, $B = H_T$, and $C = H_{S\cap T}$. The image hypothesis is exactly IP in Q. The subgroup on the left of the containment in (M2) is $K_S \cap K_T$. Its containment in $H_{S\cap T}$ is equivalent to its containment in $K_{S\cap T}$; the reverse inclusion always holds. This proves each required intersection and its converse. □

The next result concerns the opposite direction. Passing to a quotient can create new intersections, and this defect is a product condition in its kernel.

Theorem UA.Q (exact intersections of images). Let $q : H \to Q$ be a homomorphism with kernel N, and let $A, B \le H$. Then

$$q(A) \cap q(B) = q(A \cap B) \quad \Longleftrightarrow \quad N \cap AB = (N \cap A)(N \cap B). \tag{M4}$$

If q is injective on A and B, the right side reduces to $N \cap AB = \{1\}$. If $N \le A$, the equality always holds.

Proof. Suppose the image intersection is exact, and take $ab \in N$, where $a \in A$ and $b \in B$. Since $q(a) = q(b)^{-1}$, choose $c \in A \cap B$ with this image. Then

$$ab = (ac^{-1})(cb) \in (N \cap A)(N \cap B).$$

The reverse inclusion requires no hypothesis. Conversely, if $q(a) = q(b)$, then $ab^{-1} \in N \cap AB$. Write $ab^{-1} = uv$, with $u \in N \cap A$ and $v \in N \cap B$. The element $u^{-1}a = vb$ lies in $A \cap B$ and has image $q(a)$, proving the required image equality. Under injectivity both factors on the right of (M4) are trivial. If $N \le A$, both sides of that equality are N. □

The rank-three string test

Rank-three test. For three distinct involutions with commuting endpoints and adjacent products of order at least three, IP is equivalent to $G_{01} \cap G_{12} = G_1$.

Proof. Necessity is immediate. For sufficiency, the endpoint subgroup is $G_{02} = \{1, r_0, r_2, r_0r_2\}$. If $G_{02} \cap G_{01}$ contains r_2 or r_0r_2, then $r_2 \in G_{01} \cap G_{12} = G_1$, contradicting distinctness. Thus this intersection is G_0; dually $G_{02} \cap G_{12} = G_2$. These equalities exclude every generator from the subgroup on the other two. Intersections involving singletons now follow, and intersections involving the empty or full index set are immediate. This covers all subset pairs. □

3 Deletion maps and supported extensions

A deletion for $S \subseteq I$ is an endomorphism $e_S : G \to G$ satisfying $e_S(r_i) = r_i$ for $i \in S$ and $e_S(r_i) = 1$ otherwise. Requiring these values matters: a general retraction onto G_S need not be a deletion.

Proposition UA.D (Boolean deletions). If a marked group has a deletion e_S for every $S \subseteq I$, then

$$e_S e_T = e_{S \cap T}, \qquad \operatorname{im} e_S = \operatorname{Fix}(e_S) = G_S. \tag{D1}$$

The marking has IP, and

$$\ker e_S = \langle\!\langle r_i : i \notin S \rangle\!\rangle_G, \qquad G = (\ker e_S) \rtimes G_S. \tag{D2}$$

Proof. The two maps in the first equality agree on every generator. The image is generated by precisely the retained markers, and each of those markers is fixed; this proves the image and fixed-point statements. An element of $G_S \cap G_T$ is fixed by $e_S e_T = e_{S \cap T}$, so belongs to $G_{S \cap T}$. The reverse inclusion is immediate.

Let M be the normal closure in (D2). Clearly $M \le \ker e_S$. In G/M, every omitted marker is trivial, so the image of any word equals the image of its deletion. A word in $\ker e_S$ is therefore trivial in G/M, proving equality. Finally, for $g \in G$, the factorization $g = (g e_S(g)^{-1}) e_S(g)$ has factors in $\ker e_S$ and G_S. Their intersection is trivial because e_S fixes G_S. This is the asserted semidirect product. □

We now assemble extensions whose extra kernel is invisible on each proper part of its support. The groups and kernels may be infinite and nonabelian.

Theorem UA.SK (supported-kernel realization). Let I be finite and let $\mathcal{F}$ be a finite family of distinct subsets of I. Let $B = \langle b_i : i \in I \rangle$ have IP. For every $S \in \mathcal{F}$, suppose the following data are given:

1. A deletion homomorphism $p_S : B \to B_S$, fixing b_i for $i \in S$ and killing b_i otherwise.
2. A group $E_S = \langle e_{S,i} : i \in S \rangle$ and a marked epimorphism $q_S : E_S \to B_S$.
3. Injectivity of q_S on $(E_S)_T = \langle e_{S,i} : i \in T \rangle$ for every $T \subsetneq S$.

Put $N_S = \ker q_S$, and form the group

$$H = \left\{ (b, (x_S)_{S \in \mathcal{F}}) \in B \times \prod_{S \in \mathcal{F}} E_S : q_S(x_S) = p_S(b) \text{ for every } S \right\}. \tag{K1}$$

For $i \in I$, define h_i to have base coordinate b_i and S-coordinate $e_{S,i}$ when $i \in S$, and 1 otherwise. These elements generate all of H. If $\pi : H \to B$ is the base projection and $H_L = \langle h_i : i \in L \rangle$, then, for every $L \subseteq I$,

$$\pi(H_L) = B_L, \qquad \ker \pi = \prod_{S \in \mathcal{F}} N_S, \qquad \ker \pi \cap H_L = \prod_{\substack{S \in \mathcal{F} \\ S \subseteq L}} N_S. \tag{K2}$$

The factors on the right are embedded as pure coordinate subgroups. The marking of H has IP. If B and all E_S are finite, then

$$|H_L| = |B_L| \prod_{\substack{S \in \mathcal{F} \\ S \subseteq L}} |N_S|. \tag{K3}$$

Proof. The compatibility equations in (K1) are preserved by products and inverses, so H is a subgroup of the displayed direct product. The prescribed tuple h_i satisfies those equations by the literal deletion rule. Let H' be the subgroup generated by these tuples. Its base image is B, and the projection of $H'_S = \langle h_i : i \in S \rangle$ onto E_S is onto.

We first prove, by induction on inclusion among the finitely many supports, that every pure coordinate element of N_S belongs to H'_S. Fix $n \in N_S$. Choose a word on the markers indexed by S which evaluates to n in E_S, and evaluate the same word in H', obtaining $w \in H'_S$. Its base coordinate lies in B_S and is $q_S(n) = 1$.

Consider its coordinate at another support T. Because the word uses only markers from S, that coordinate lies in $(E_T)_{S \cap T}$. Its image under q_T is $p_T(1) = 1$. If $T \not\subseteq S$, then $S \cap T \subsetneq T$, and the proper-parabolic injectivity hypothesis forces this coordinate to be 1. The only possible nonidentity coordinates of w are consequently at supports $T \subseteq S$. Its S-coordinate is the prescribed n, while every coordinate at a proper support $T \subsetneq S$ belongs to N_T.

For each such proper support, the induction hypothesis supplies the pure coordinate element having exactly that value in N_T, as an element of $H'_T \leq H'_S$. Multiply w by the inverses of those elements. Distinct pure coordinate subgroups commute in the ambient direct product, so these operations remove the unwanted coordinates without changing the S-coordinate. The result is the pure element n in coordinate S, and remains in H'_S. A minimal support has no smaller coordinates to remove, so this argument includes the initial step. No commutativity within an individual kernel has been used.

The kernel of π in the full group H is exactly the direct product of the N_S, by (K1). The induction places that entire kernel in H'. Since H' already maps onto B, every $h \in H$ has the same base coordinate as some $h' \in H'$; then $h(h')^{-1} \in \ker \pi \leq H'$. Thus $H' = H$.

The image equality $\pi(H_L) = B_L$ follows on generators. If $x \in H_L \cap \ker \pi$, its S-coordinate lies in $(E_S)_{L \cap S}$ and in N_S. This coordinate is trivial whenever $S \not\subseteq L$, again by proper-parabolic injectivity. Conversely, when $S \subseteq L$, the isolation argument placed the entire pure N_S in $H_S \leq H_L$. This proves all three assertions in (K2).

The coordinate products in (K2) satisfy

$$\left(\prod_{S \subseteq L} N_S \right) \cap \left(\prod_{S \subseteq M} N_S \right) = \prod_{S \subseteq L \cap M} N_S.$$

Theorem UA.K now proves IP in H. More explicitly, an element of $H_L \cap H_M$ has base in $B_{L\cap M}$; subtracting a lift from $H_{L\cap M}$ leaves an element in both displayed kernel products, hence in the kernel product for $L \cap M$. This also shows exactly where IP of the base is used. Finally (K3) is the order formula for the surjection $H_L \to B_L$ with the kernel just computed. □

Corollary UA.SK.Local (recognition on an antichain). Under UA.SK, if $\mathcal{F}$ is an antichain under inclusion, then the coordinate projection $H_S \to E_S$ is a marked isomorphism for every $S \in \mathcal{F}$.

Proof. It is onto on generators. If an element maps to 1, its base coordinate also vanishes, since the base value of a word on S is the q_S-image of its S-coordinate. Formula (K2) says that the remaining kernel in H_S has only the factor N_S: the antichain excludes every other support contained in S. The S-coordinate detects that factor, so the element is 1. □

Corollary UA.SK.Delete (derived deletions). If, in addition, B has deletions for every subset of I, then every E_S and H has all its Boolean deletions.

Proof. For $T \subsetneq S$, the restriction $q_S : (E_S)_T \to B_T$ is a marked isomorphism: it is onto on generators and injective by hypothesis. Compose q_S, the deletion $B_S \to B_T$, and the inverse of this isomorphism. This gives a homomorphism $E_S \to (E_S)_T$ that fixes exactly the T-markers and kills the others. For $T = S$ use the identity. Proposition UA.D supplies the Boolean identities.

For $L \subseteq I$, apply the base deletion e_L and, in coordinate S, the local deletion onto $S \cap L$. The two sides of each compatibility equation agree on the local markers; the base deletions commute by UA.D. Thus the coordinatewise operation preserves (K1). On h_i it fixes h_i if $i \in L$, and kills it otherwise. It is the required deletion of H. □

Corollary UA.SK.Central (lower-central kernel). Under UA.SK, suppose $c \geq 1$, $\gamma_c(B) = 1$, and $N_S = \gamma_c(E_S)$ for every support. Then

$$\gamma_c(H) = \ker \pi. \tag{K4}$$

Proof. Applying π to a commutator of weight c gives 1, so $\gamma_c(H) \leq \ker \pi$. For the converse, repeat the inclusion induction in the proof of UA.SK. The projection $H_S \to E_S$ is onto; therefore its image on $\gamma_c(H_S)$ is $\gamma_c(E_S) = N_S$. Given $n \in N_S$, choose a lift $w \in \gamma_c(H_S)$. Its base is 1, since $\gamma_c(B) = 1$. The same proper-support argument makes all coordinates at $T \not\subseteq S$ trivial. The remaining proper-support coordinates lie in N_T; by induction their pure copies belong to $\gamma_c(H)$. Removing them leaves the pure n, still in $\gamma_c(H)$. All pure kernels are obtained, proving (K4). □

Corollary UA.SK.Separator (factorization across a separator). Under UA.SK, let $I = L \cup M$. Suppose $B = B_L B_M$ as a product of sets, and every support in $\mathcal{F}$ is contained in L or in M. Then $H = H_L H_M$.

Proof. For $h \in H$, factor $\pi(h) = b_L b_M$ and choose lifts $u_L \in H_L$, $u_M \in H_M$. Then $h = n u_L u_M$ for some $n \in \ker \pi$. Assign every pure coordinate factor of n to one of the two sides containing its support. Formula (K2) gives $n = n_L n_M$, with $n_L \in H_L$, $n_M \in H_M$. Each pure N_S is normal in H, because conjugation in the ambient product acts coordinatewise and $N_S \trianglelefteq E_S$. Hence the product of the assigned M-factors is normal in H. Moving n_M past u_L gives

$$h = (n_L u_L)(u_L^{-1} n_M u_L \, u_M) \in H_L H_M.$$

The opposite inclusion is automatic. □

The hypotheses of UA.SK do not identify H with a universal amalgam of the E_S. Additional cross-relations and a presentation-recognition argument may be necessary. Nor can injectivity on every proper parabolic be replaced by injectivity on individual markers alone.

Proposition UA.Fiber (two compatible split factors). Let A and B be marked groups on finite label sets L and M, and put $J = L \cap M$. Suppose their J-parabolics are identified with the same marked group C, and that $p_A : A \to C$ and $p_B : B \to C$ are deletions. Then

$$A \times_C B \cong (\ker p_A \times \ker p_B) \rtimes C. \tag{F1}$$

This full fiber product is generated by the tuples $(a_i, 1)$ for $i \in L \setminus J$, $(1, b_i)$ for $i \in M \setminus J$, and (c_i, c_i) for $i \in J$. The two original groups embed through $a \mapsto (a, p_A(a))$ and $b \mapsto (p_B(b), b)$. If finite, its order is $|A||B|/|C|$.

Proof. Every compatible pair has a unique form $(n_A c, n_B c)$, giving the split decomposition and the order formula. The stated copies of A and B are generated by subsets of the stated tuple marking. They contain $(n_A, 1)$ and $(1, n_B)$, respectively, while either contains the diagonal copy of C. These subgroups generate the split product, proving the assertion. The embedding formulas are injective because their first and second projections, respectively, are identities. □

4 Recognition of presentations and compatible products

All homomorphisms in this paper are group homomorphisms. A product of subgroups means a product of sets until it has been shown to be a subgroup. No group is assumed finite in this section.

Theorem UA.Normal (normal kernel recognition). Let $\pi : P \to B$ and $\rho_i : P \to E_i$, for i in a finite set J, be homomorphisms. Suppose $K_i \trianglelefteq P$ satisfy

$$\ker \pi = \langle K_i : i \in J \rangle, \tag{1}$$

$$\rho_i(K_j) = 1 \quad (i \neq j), \tag{2}$$

$$\ker \rho_i \cap K_i = 1. \tag{3}$$

Then the homomorphism

$$P \longrightarrow B \times \prod_{i \in J} E_i, \qquad x \longmapsto (\pi(x), (\rho_i(x))_i)$$

is injective. Moreover, the kernel in (1) is the internal direct product of the K_i. For injectivity alone it suffices to replace (1) by $\ker \pi \leq \langle K_i : i \in J \rangle$.

Proof. For $i \neq j$, an element of $K_i \cap K_j$ is killed by ρ_i by (2), and is therefore trivial by (3). Normality gives $[K_i, K_j] \leq K_i \cap K_j$, so distinct factors commute. Their finite set product is consequently their generated subgroup. If $\prod_i k_i = 1$ with $k_i \in K_i$, applying ρ_j gives $\rho_j(k_j) = 1$, whence $k_j = 1$. Thus multiplication identifies the external product with the generated subgroup. Finally, an element in the kernel of the displayed homomorphism belongs to $\ker \pi$, hence has such a product expression. Every coordinate vanishes, so every factor is trivial. □

Corollary UA.RecognizeQuotient. Let $f : P \twoheadrightarrow H$ be a homomorphism, and let H have a base homomorphism $p : H \to B$ and coordinates $c_i : H \to E_i$. If UA.Normal applies to pf, $c_i f$ and normal subgroups $K_i \leq P$, then f is an isomorphism.

Proof. The coordinate map factors through f, so injectivity of the coordinate map implies injectivity of f. Surjectivity is the stated hypothesis. □

Normality is substantial. Let $P = D_8 = \langle u, v \mid u^2 = v^2 = (uv)^4 = 1 \rangle$, let $B = 1$, and take $K_1 = \langle u \rangle$, $K_2 = \langle v \rangle$. Its abelianization $P \to C_2^2$ has coordinates faithfully detecting the two factors and killing the other one. The factors generate P, but the map has the nontrivial kernel $\langle (uv)^2 \rangle$. Here neither factor is normal. Thus a correct local model and coordinate detection do not themselves eliminate mixed relations.

The following specialization uses the full compatible product H in UA.SK and assumes that its support family is an antichain.

Theorem UA.Recognition (complete presentation). In this setting, let $P = \langle r_t : t \in I \rangle$ be given by a presentation. Suppose:

1. its relators hold on the marker tuples in the compatible product H;
2. the local relations of every E_S hold on $(r_t)_{t \in S}$;
3. the image K_S of N_S under $E_S \to P$ is normal in P;

4. adjoining the relations $K_S = 1$ for all S gives exactly the marked presentation of B.

Then the marker map $P \to H$ is an isomorphism. In particular the K_S form an internal direct product, all marked intersections are exact, and if the input groups are finite then

$$|P_L| = |B_L| \prod_{S \subseteq L} |N_S|. \tag{4}$$

The additional deletion conclusion of UA.SK.Delete transfers to P under its hypotheses. The same holds for the geometric conclusion of UA.SK.Geometry under its hypotheses.

Proof. The first assumption and the generation assertion of UA.SK give a marked surjection $f : P \twoheadrightarrow H$. The second gives $E_S \to P_S$; its composite with f and projection to E_S is the identity on every local marker, hence is the identity homomorphism. Thus each local map is injective, without an order comparison. Under f, the image of N_S is its pure S-coordinate: the antichain makes every other intersection of supports proper, where q_T is injective. Consequently the coordinate maps kill K_S outside its own coordinate and are injective on K_S in that coordinate.

Normality makes $K = \langle K_S : S \in \mathcal{S} \rangle$ a normal subgroup. The last assumption identifies the marked map $P/K \to B$ with an isomorphism, so the base map $P \to B$ has kernel exactly K. UA.RecognizeQuotient now proves that f is an isomorphism. The remaining conclusions follow from UA.SK and the explicitly indicated additional clauses. □

The last assumption is a presentation calculation. It does not assume the isomorphism $P \cong H$ which is being proved. In practice it says that, after killing the displayed local kernel words, the remaining relations are the known base presentation. A pair of marked maps which are inverse on generators also proves this condition. Normality likewise has a concrete local test: the markers within S already normalize K_S, and one must check the remaining markers. It is enough that those remaining markers centralize a generating set of K_S. Merely writing down the desired quotient does not supply this check.

Proposition UA.Mix (jointly faithful deletions). Let $G = \langle r_i : i \in I \rangle$ have every Boolean deletion e_J. Suppose $I = L \cup M$, put $C = L \cap M$, and assume $\ker e_L \cap \ker e_M = 1$. Then

$$G \longrightarrow G_L \times_{G_C} G_M, \qquad g \longmapsto (e_L(g), e_M(g))$$

is a marked isomorphism. The maps to G_C on the right are the restrictions of the corresponding deletions. No group is assumed finite.

Proof. UA.D gives $e_L e_M = e_C = e_M e_L$. Hence the displayed pair is compatible, and the homomorphism's kernel is the stated trivial intersection. For a compatible pair $(x, y) \in G_L \times_{G_C} G_M$, put $c = e_C(x) = e_C(y)$. Since the two deletions fix their respective parabolics, $e_M(x) = c = e_L(y)$ and both deletions fix c. Thus

$$e_L(xc^{-1}y) = xc^{-1}c = x, \qquad e_M(xc^{-1}y) = cc^{-1}y = y.$$

Every compatible pair therefore has the explicit preimage $xc^{-1}y$. The homomorphism is surjective as well as injective, with inverse given by this formula. Its values on the markers give the stated marking. □

The kernel condition can be checked by coordinates. In a supported-kernel realization with inherited deletions, suppose every base coordinate is detected on at least one side, and every support is contained in L or M. An element killed by both deletions has trivial base coordinate. Each pure supported-kernel coordinate is then detected on a side containing its support, so all these coordinates are trivial too. This proves the kernel condition in the path and two-branch applications below; it is not assumed from a coincidence of finite orders.

5 Coordinate kernels and incidence geometries

An incidence system has a finite type set, objects carrying types, and a symmetric incidence relation; distinct objects of one type are not incident. A flag is a set of pairwise incident objects. A chamber has one object of each type. A geometry is an incidence system in which every flag extends to a chamber. The residue of a flag consists of the objects of the remaining types incident with that flag, with induced incidence. The geometry is residually connected when the incidence graph of every residue of rank at least two, including the empty-flag residue, is connected. It is thin when every panel, a flag missing just one type, lies in exactly two chambers. A regular hypertope is a thin, residually connected geometry admitting a chamber-transitive type-preserving automorphism group.

For a marked group indexed by I, its maximal parabolics are $P_i = G_{I\setminus\{i\}}$. The associated typed coset incidence system has objects $P_i g$, and incidence means nonempty intersection of cosets. All actions and cosets below use this right-coset convention.

For subgroups $P_i \leq G$, typed coset Helly means that every pairwise intersecting family of right cosets $P_i g_i$, with distinct indices, has a common element. This property is separate from IP. It supplies a common representative for an incident flag, whereas IP determines the stabilizer of a flag already represented in this way.

Proposition UA.Geometry (coset recognition and faithful chamber action). Suppose the marking of G consists of nonidentity involutions, has IP, and has typed coset Helly for its maximal parabolics. The coset incidence system is then a thin, residually connected geometry, and G is its full type-preserving automorphism group, acting regularly on chambers. More generally, the full type-preserving automorphism group of any thin, residually connected geometry of finite rank acts freely on chambers. A chamber-transitive subgroup is consequently the full group.

Proof. In the coset system, a common coset point extends every flag to a chamber by choosing the coset of each missing type through that point. Right translation sends every such flag to its base flag. For a base flag of type set S, IP gives stabilizer $G_{I\setminus S}$. In particular the full chamber stabilizer is 1, while a panel stabilizer is a group generated by one nonidentity involution and has order two. There are therefore exactly two chambers through each panel.

Consider a residue with remaining type set T, where $|T| \geq 2$. Its stabilizer is G_T, and its maximal object stabilizers are $G_{T\setminus\{i\}}$, $i \in T$. Every marker indexed by T lies in at least one of these proper parabolics. A word in those subgroups supplies a walk between their translated base objects in the residue incidence graph; all objects are translates of base objects and all base objects lie in a common flag. The residue is therefore connected.

For the general assertion, first observe that every residually connected finite-rank geometry has a connected chamber graph. This follows by induction on rank: chambers through a fixed object are connected through galleries in its residue, and a path in the incidence graph, extending each consecutive incident pair to a chamber, connects arbitrary chambers through these subgraphs. The rank-one case holds because all chambers share the empty panel. In a thin geometry an automorphism fixing a chamber fixes its unique neighbor across each panel. It therefore fixes every chamber along every gallery, and hence every object. The chamber action is free. If a subgroup is chamber-transitive, any full automorphism agrees on a base chamber with one subgroup element; their quotient fixes that chamber and is the identity. Apply this to the coset geometry to obtain the stated full automorphism group. □

5.1 Helly transfer through coordinate kernels

Write right cosets as Pg, with the subgroup on the left. Let $q : H \to B$ be an epimorphism of groups generated by corresponding marked families $(h_i)_{i\in I}$ and $(b_i)_{i\in I}$, where I is finite. For $J \subseteq I$, write $H_J = \langle h_j : j \in J\rangle$ and $B_J = \langle b_j : j \in J\rangle$. In particular, $q(H_J) = B_J$ for every J.

Fix any set $\mathcal{J}$ of subsets of I as the permitted types. The maximal-parabolic case is $\mathcal{J} = \{I \setminus \{i\} : i \in I\}$.

Theorem UA.SK.Helly (coordinate-kernel transfer). Suppose B has IP. Suppose $N = \ker q$ is an internal direct product $\prod_{\alpha\in A} N_\alpha$, with a finite factor set A, and for every $J \in \mathcal{J}$,

$$N \cap H_J = \prod_{\alpha\in A_J} N_\alpha \qquad \text{for some } A_J \subseteq A. \tag{H1}$$

If the family $(B_J)_{J\in\mathcal{J}}$ has typed coset Helly, then so does $(H_J)_{J\in\mathcal{J}}$.

No commutativity within the factors, finiteness of H or B, or faithfulness of an action of B on N is assumed. The required intersection-image fact is $q(H_{J\cap K}) = B_J \cap B_K$; it follows here from the marked map and IP of B. IP of H is not an additional assumption for this transfer theorem.

Proof. Consider pairwise intersecting cosets $H_{J_i}g_i$. Their images $B_{J_i}q(g_i)$ intersect pairwise. Choose a common base element b by the assumed Helly property, and choose u in H with $q(u) = b$. Each set

$$F_i = H_{J_i}g_i \cap q^{-1}(b)$$

is nonempty: because $bq(g_i)^{-1} \in B_{J_i}$, a preimage of that element in H_{J_i}, multiplied on the right by g_i, belongs to F_i.

We first show that $F_i \cap F_j \neq \emptyset$. Choose $x \in H_{J_i}g_i \cap H_{J_j}g_j$ and put z=q(x). Both b and z belong to each of the two base cosets, so

$$bz^{-1} \in B_{J_i} \cap B_{J_j} = B_{J_i\cap J_j}.$$

The marked map is surjective on this intersection parabolic. Choose $h \in H_{J_i\cap J_j}$ with $q(h) = bz^{-1}$. Since h belongs to both H_{J_i} and H_{J_j}, left multiplication by h preserves both right cosets. Thus hx belongs to their intersection and has image b. This proves the assertion without assuming that q carries an arbitrary intersection onto the intersection of its images.

Choose $x_i \in F_i$ and put $k_i = x_iu^{-1} \in N$. If $y \in F_i$, then $yx_i^{-1} \in N \cap H_{J_i}$; the converse also holds. Hence

$$F_iu^{-1} = (N \cap H_{J_i})k_i. \tag{H2}$$

Under the direct product coordinates of N, (H1) makes (H2) unrestricted in coordinates $\alpha \in A_{J_i}$, and fixes coordinate $\alpha \notin A_{J_i}$ to $(k_i)_\alpha$. Pairwise intersections imply that any two constraints fixing the same coordinate fix it to the same value. Choose that value in every constrained coordinate and the identity in every unconstrained coordinate. The resulting element belongs to all sets in (H2). Multiplication on the right by u supplies a common element of the original cosets. This proves the theorem. □

Corollary UA.SK.Geometry. In the supported-kernel realization UA.SK, assume the base marking gives a regular hypertope and every marked generator in every attached group is an involution. Then the resulting marking also gives a regular hypertope, and its full type-preserving automorphism group is the constructed group.

Proof. UA.SK gives IP and the exact profile

$$N \cap H_J = \prod_{S\subseteq J} N_S.$$

The base maximal-parabolic cosets have typed Helly. Theorem UA.SK.Helly applies with factors indexed by the supports S. Each constructed generator is a tuple whose base coordinate and every nontrivial attached coordinate are involutions; hence its square is the identity. Its base image is nonidentity, so its order is exactly two. UA.Geometry now gives thinness, residual connectedness, chamber regularity and equality with the full type-preserving automorphism group. In particular, this conclusion uses typed Helly in addition to IP. □

A sufficient alternative to local involutivity is that every support has size at least two: each singleton local parabolic is then proper, and injectivity into its nonidentity involutory base image forces the local marker to have order two. This is a sufficient condition, not an equivalent reformulation. Without either condition the geometric corollary fails. With a singleton type set and support, the quotient

$C_4 \to C_2$ is injective on its only proper parabolic, the trivial subgroup. UA.SK then constructs C_4, whose distinguished generator has order four. In rank three, adjoining two independent C_2 markers gives $C_4 \times C_2^2$ with the same defective singleton attachment and a four-element panel.

6 Frattini coordinates

Theorem UA.Phi (primitive saturation and coordinates). Let P be a finite p-group with an irredundant generating marking $(r_i)_{i\in I}$, $|I| = d$, and let $F = \Phi(P)$. Then, for every $S \subseteq I$,

$$\Phi(P_S) = P_S \cap F, \qquad P_S F = \pi^{-1}\operatorname{span}(\bar{r}_i : i \in S), \qquad [P : P_S F] = p^{d-|S|}. \tag{P1}$$

The saturated parabolics $P_S F$ intersect according to intersection of index sets. Each omission subgroup $P_{I\setminus\{i\}}$ has exactly one maximal overgroup, namely $P_{I\setminus\{i\}}F$.

If $B \geq F$ and $(s_i)_{i\in J}$ is an irredundant relative generating family over B, then $B \trianglelefteq P$, P/B has basis $(s_i B)_{i\in J}$ over $\mathbb{F}_p$, and

$$\langle B, s_i : i \in T\rangle/B = \operatorname{span}(s_i B : i \in T). \tag{P2}$$

If $p = 2$ and all original r_i are involutions, then $F = P'$ and $\Phi(P_S) = P_S'$ for every S.

Proof. Burnside's basis theorem makes the $\bar{r}_i$ a basis of P/F. The image of P_S therefore has order $p^{|S|}$, with kernel $P_S \cap F$. Since this image is elementary abelian, $\Phi(P_S) \leq P_S \cap F$. On the other hand P_S is generated by $|S|$ elements, so Burnside gives $[P_S : \Phi(P_S)] \leq p^{|S|}$. The quotient by $P_S \cap F$ already has that order. Equality follows, proving the first formula. The remaining formulas and the intersection assertion are inverse-image statements for coordinate subspaces.

Every maximal subgroup of a finite p-group contains F and corresponds to a hyperplane of P/F. A hyperplane containing a specified coordinate hyperplane equals it, proving uniqueness of the maximal overgroup of an omission subgroup.

For the relative assertion, every subgroup containing F is normal, because P/F is abelian. The quotient P/B is elementary abelian. The indicated images span it, and any nontrivial linear relation would express one s_i using B and the other markers, contradicting irredundancy. They form a basis, proving (P2). Finally, an abelian group generated by involutions is elementary abelian of exponent two. Thus $P' \leq \Phi(P)$, while the elementary abelian quotient P/P' gives the reverse inclusion. The same argument applies to each P_S. □

These coordinates also describe the corresponding omission cosets: a coset prescribes one coordinate value, and every simultaneous assignment gives a chamber. The geometry is complete multipartite, with p objects in each type, and the quotient translation group acts regularly on chambers. For $p > 2$ it need not be the full type-preserving automorphism group; the thin case $p = 2$ is the one covered by UA.Geometry.

7 The canonical quotient of a forest marking

Suppose a group is generated by nonidentity involutions r_i, $i \in I$. Its noncommutation graph has vertex set I, and an edge $\{i, j\}$ precisely when r_i and r_j do not commute. We write $G_J = \langle r_j : j \in J\rangle$, as in the marking definitions. The intersection property in this section always refers to the original group. It is not assumed to survive an intermediate quotient.

Lemma UA.Lift (descending lift). Let G be a finite nilpotent group, let $A, B \leq G$ satisfy $A \cap B = 1$, and fix an integer $m \geq 1$. Put $D_j = \gamma_j(G)$, $A_j = A \cap D_j$, $B_j = B \cap D_j$. If the images of A_j, B_j generate D_j/D_{j+1} for every $j \geq m$, then $D_j = A_j B_j$ for every $j \geq m$.

Proof. Descend from a trivial term. Assuming $D_{j+1} = A_{j+1}B_{j+1}$, if $a \in A_j, b \in B_j$ have equal images in the next quotient, write $ab^{-1} = a'b'$ with $a' \in A_{j+1}, b' \in B_{j+1}$. Then $a'^{-1}a = b'b \in A \cap B$, so both images are trivial. Their images therefore form an internal direct sum in the abelian quotient.

Counting gives $|D_j| = |A_j||B_j| = |A_jB_j|$, proving equality as sets. No prior normality or permutability of A, B is assumed. ■

Lemma UA.Support (graded support). Let a positively graded Lie algebra be generated in degree one by disjoint sets X, Y, with subalgebras L_X, L_Y. If $[x, y] \in L_Y$ for $x \in X, y \in Y$, then $L = L_X + L_Y$, degree by degree.

Proof. The derivation $\operatorname{ad} x$ preserves L_Y because it does so on its generators. Iterated brackets of those derivations also preserve L_Y, by Jacobi. Hence $[L_X, L_Y] \subseteq L_Y$, and $L_X + L_Y$ is a Lie subalgebra containing all generators. The argument holds in characteristic two. ■

Theorem UA.Forest (forest quotient). Suppose G is a finite two-group generated by nonidentity involutions r_i, with marked IP, and its noncommuting pairs form a forest T on d vertices and m edges. Then edge commutators are independent in $G'/\gamma_3(G)$, and $G/\gamma_3(G)$ is the marked group

$$(v, z)(w, t) = (v + w, z + t + \beta(v, w)), \quad v, w \in \mathbb{F}_2^d,\ z, t \in \mathbb{F}_2^m, \tag{F0}$$

where, after orienting each edge, its component of β is $v_{\text{head}} w_{\text{tail}}$. Its order is 2^{d+m}.

Proof. IP gives irredundancy. UA.Phi gives $G/G' = C_2^d$ and $G' \cap \langle r_i \rangle = 1$. The lower-central graded Lie ring is generated by these degree-one elements; all its graded components are $\mathbb{F}_2$-spaces by bilinearity. Edge commutators span degree two. If they satisfy a nonzero linear relation, root each forest component and choose a participating edge whose child endpoint has maximal depth in its component. Removing it separates the descendant set X from all other vertices Y, and no other participating edge is internal to X. The relation therefore writes the only nonzero cross bracket as a sum of brackets internal to Y. Lemma UA.Support shows every graded component is generated by pure X- and pure Y-brackets. Their representatives lie in $G_X \cap \gamma_j(G)$ and $G_Y \cap \gamma_j(G)$.

Actual IP gives $G_X \cap G_Y = 1$, so Lemma UA.Lift gives $G' = (G' \cap G_X)(G' \cap G_Y)$. For the crossing endpoints $x \in X, y \in Y$, write $[r_x, r_y] = uv$ accordingly. Then $u \in G_X \cap G_{\{x\} \cup Y} = \langle r_x \rangle$, whence $u = 1$; next $[r_x, r_y] \in G_Y \cap G_{\{x,y\}} = \langle r_y \rangle$, so it too is one. This contradicts that the edge is noncommuting. Independence follows.

Bilinearity makes (F0) associative; its coordinate markers are involutions with independent central edge commutators. It is generated by those markers, and its parabolic on any subset J consists exactly of vertex coordinates supported on J and edge coordinates supported on edges internal to J. Thus it has IP. Restricting the vertex coordinates to a subset and retaining exactly its internal edge coordinates preserves the multiplication formula, so it also gives every Boolean deletion. Collection in the corresponding class-two presentation gives at most 2^{d+m} elements, and this coordinate model attains the bound. Independence in $G'/\gamma_3(G)$ gives the same order for $G/\gamma_3(G)$; hence the marked map is an isomorphism. ■

8 Finite quotients of prescribed excess order

The next theorem separates a universal finite search space from the task of proving that a particular family has the required class-two quotient. That identification is an application hypothesis, not a consequence of the search.

Theorem UA.Nilpotent (complete marked kernel parameter space). Let $F = \langle x_i : i \in I \rangle$, where $|I| = d$ and $x_i^2 = 1$. Suppose F has all Boolean deletions. Put $Q = F/\gamma_3(F)$, with its image marking. For $k \geq 0$, set

$$U = F/\gamma_{k+3}(F), \qquad K = \gamma_3(U), \qquad K_S = K \cap U_S. \tag{N1}$$

Then U is a finite two-group,

$$|U| \leq 2^{\sum_{j=1}^{k+2} d^j}, \tag{N2}$$

the markings of U and Q have Boolean deletions and IP, and U/K is canonically the marked group Q. For every $S \subseteq I$,

$$K_S = \gamma_3(U_S). \tag{N3}$$

There is a bijection between the following two sets:

1. Ordered marked isomorphism classes of finite two-group quotients $f : F \twoheadrightarrow G$ for which the induced marked map $Q \to G/\gamma_3(G)$ is an isomorphism, $|G| = |Q|2^k$, and G has IP.
2. Normal subgroups $N \trianglelefteq U$ satisfying $N \le K$, $[K : N] = 2^k$, and

$$\frac{NK_S}{N} \cap \frac{NK_T}{N} = \frac{NK_{S\cap T}}{N} \quad (S, T \subseteq I). \tag{N4}$$

The correspondence sends N to U/N with the original image markers. If $D \le \mathrm{Sym}(I)$ is specified and each $\sigma \in D$ induces an automorphism $x_i \mapsto x_{\sigma(i)}$ of F, then equivalence allowing precisely these label permutations is the D-orbit relation on the kernels in item 2.

Proof. The degree-one part of the lower-central graded Lie ring of F is generated by elements of exponent dividing two. By multilinearity, every iterated commutator in those generators also has exponent dividing two in its homogeneous component. In degree j, at most d^j left-normed commutators suffice to generate that component. Consequently

$$|\gamma_j(F)/\gamma_{j+1}(F)| \le 2^{d^j}.$$

Multiplying these bounds for $1 \le j \le k+2$ proves finiteness, the two-power order and (N2). Each deletion preserves every lower-central term, and therefore descends both to U and to Q. Proposition UA.D gives IP for both image markings; no separate IP hypothesis on Q is needed. The epimorphic-image property of the lower central series gives $U/K \cong F/\gamma_3(F) = Q$.

For (N3), one inclusion follows from $U_S \le U$. If $z \in \gamma_3(U) \cap U_S$, the deletion e_S fixes z and sends $\gamma_3(U)$ into $\gamma_3(U_S)$. This proves the reverse inclusion.

Let G satisfy item 1. Its subgroup $\gamma_3(G)$ has order 2^k. In a nilpotent group, equality between two successive nontrivial lower-central terms would persist at every later term, contradicting termination of that series. Thus each nontrivial step beginning at $\gamma_3(G)$ strictly decreases its order. There are at most k such steps, and $\gamma_{k+3}(G) = 1$. The map $F \to G$ therefore factors through U. Its kernel N lies in K, since the induced map $U/K \to G/\gamma_3(G)$ is an isomorphism, and the orders give $[K : N] = 2^k$.

Conversely, let $N \trianglelefteq U$ with $N \le K$ and $[K : N] = 2^k$, and write $G = U/N$. Its third lower-central subgroup is K/N, its canonical class-two quotient is Q, and its order is $|Q|2^k$. If $q : U \to G$, then

$$(K/N) \cap G_S = q(K \cap U_S) = NK_S/N. \tag{N5}$$

Indeed, $q(u) \in K/N$ with $u \in U_S$ implies $u \in K$, because $q^{-1}(K/N) = K$. The reverse inclusion is immediate. Applying UA.K to $G \to Q$ proves that IP in G is equivalent to precisely (N4).

An isomorphism preserving each ordered marker commutes with the two maps from U, because it agrees with them on a generating family. Their kernels are consequently equal. Equal kernels plainly give the same marked quotient. Finally, every specified label permutation acts on F, preserves its characteristic lower-central terms, and hence acts on U and on the displayed kernels. A marker-permuting isomorphism induces the corresponding kernel equality after this automorphism, and the converse descends that automorphism to the quotients. This proves the orbit assertion at exactly the specified equivalence scope. □

Theorem UA.Ladder (invariant index-two descent). Let U be a finite two-group and $M \trianglelefteq U$. The U-normal subgroups $L \le M$ of index two are exactly the inverse images of the hyperplanes of

$$M/\big(\Phi(M)[M, U]\big). \tag{N6}$$

For $K \trianglelefteq U$, every U-normal subgroup $N \le K$ of index 2^k is obtained by k successive such steps.

Proof. An index-two subgroup L of M contains $\Phi(M)$ and has quotient C_2. If L is U-normal, conjugation by U on M/L is trivial, since $\mathrm{Aut}(C_2) = 1$. Thus $[M, U] \le L$. Conversely, a hyperplane inverse image has index two and contains $[M, U]$; for $l \in L$ and $u \in U$, the conjugate of l differs from

l by an element of $[M, U]$, so lies in L. The quotient in (N6) is elementary abelian by Burnside's basis theorem, proving the first assertion.

Suppose $N \trianglelefteq U$ and $N < M$. The nontrivial finite two-group M/N has a nonzero Frattini quotient V. Conjugation by U/N acts on the nonzero linear maps $V \to \mathbb{F}_2$. If $\dim V = a$, there are $2^a - 1$ such maps, an odd number. Every orbit has two-power size, so at least one orbit has size one. The kernel of a fixed nonzero map lifts to a U-normal subgroup between N and M, of index two in M. Repeating with the new subgroup gives a chain terminating at N; its length is k, since the total index is 2^k. □

Lemma UA.Order (presentation recognition by a sharp bound). Let $A_1, \dots, A_t$ be finite normal subgroups of G, with $|A_j| \le a_j$. Suppose $[G : A_1 \cdots A_t] \le b$. Then $|G| \le b \prod_j a_j$. If a finite quotient of G has this latter order, its quotient map is an isomorphism.

Proof. Normality makes each partial product a subgroup. The multiplication map from $A_1 \times \cdots \times A_t$ onto the product is surjective as a map of sets, so that product has order at most $\prod_j a_j$. Multiplication by the quotient index proves the bound and finiteness of G. A quotient attaining the bound forces equality of the finite orders and hence a trivial kernel. □

Corollary UA.Search (rank and excess classify the finite search). Let T be a specified forest on d vertices and m edges, and let k≥0. Set

$$F_T = \langle x_i \mid x_i^2 = 1,\ [x_i, x_j] = 1 \text{ for nonedges} \rangle, \quad U_{T,k} = F_T/\gamma_{k+3}(F_T), \quad K = \gamma_3(U_{T,k}).$$

The ordered IP markings by nonidentity involutions with graph T and order 2^{d+m+k} are exactly the quotients $U_{T,k}/N$, where N is normal, contained in K, $[K : N] = 2^k$, and the images $(N(K \cap (U_{T,k})_J))/N$ have IP for every pair J,L. Generator-set equivalence is exactly the action of Aut(T) on these kernels.

Proof. Killing any set of generators preserves every defining relator of F_T. Collection in its class-two quotient gives at most d+m binary coordinates, and (F0) realizes all of them, so this quotient is precisely the marked model Q_T. Every target has that same quotient by UA.Forest. These are exactly UA.Nilpotent's hypotheses. Its all-parabolic condition and UA.Ladder give the stated bijection and exhaustive search. An ordered isomorphism must commute with the generating map, hence has the same kernel. A permutation of the generator set preserves the noncommutation graph and extends to an automorphism of F_T and $U_{T,k}$; this proves the last claim. ■

The last corollary is a finite classification procedure, with the marking retained. First form the finite group $U_{T,k}$; then iterate UA.Ladder inside $\gamma_3(U_{T,k})$, test the explicit intersection condition of UA.Nilpotent, and take the declared graph-automorphism orbits. The theorem proves that this procedure terminates and misses no target marking. Any reported numbers of groups, named presentations or universal orders still require execution and an application-specific identification. No numerical census is assumed or reported in this paper.

9 Affine square groups

Let

$$a(x, y) = (-x, y), \quad b(x, y) = (y, x), \quad c(x, y) = (x, 1 - y).$$

Proposition UA.Affine (the complete square presentation). These generate $W = \mathbb{Z}^2 \rtimes D_8$, of type $\{4, 4\}$, with $ac = ca$. Its presentation is

$$W = \langle a, b, c \mid a^2 = b^2 = c^2 = (ab)^4 = (bc)^4 = (ac)^2 = 1 \rangle. \tag{S1}$$

Proof. Put $d = bab$, $y_0 = cd$, and $x_0 = by_0b$. The relations give $[a, d] = 1$, $[a, y_0] = 1$, $dy_0d = y_0^{-1}$, $ax_0a = x_0^{-1}$, and conjugation by b swaps x_0, y_0. Substitution of $bc = x_0ab$ makes $(bc)^4 = 1$ equivalent to $[x_0, y_0] = 1$. Thus every word collects to $x_0^u y_0^v M$, $M \in \langle a, b \rangle = D_8$. In the affine model x_0, y_0 are the unit translations, so these forms are distinct. This proves (S1). □

The normal form also fixes our affine multiplication convention: the rightmost transformation acts first. The symbol D_8 denotes the group of signed permutation matrices in two coordinates, of order eight. In all dihedral notation below the subscript denotes the order of the group.

Proposition UA.Square (finite square seeds). For $n \geq 5$, define the invariant lattice

$$\Lambda_n = \begin{cases} 2^e\mathbb{Z}^2, & n = 2e+3, \\ 2^e\langle (1,1),(1,-1)\rangle, & n = 2e+4, \end{cases} \qquad e \geq 1,$$

and put $H_n = W/\Lambda_n$. Its order is 2^n; the full linear D_8 coordinate is retained, even if its action on the vertex quotient is not faithful. Its pair orders are $4, 4, 2$. It has all Boolean deletions and IP. Its third lower-central subgroup is $2\mathbb{Z}^2/\Lambda_n$, of order 2^{n-5}.

Proof. The lattice indices are respectively 2^{2e} and 2^{2e+1}, so the normal form gives the stated orders. The three linear pair products have orders four, four and two; hence the marked pair orders cannot fall in either lattice quotient.

Write $T = abcb$, $U = cbab$, translations by $(-1, 0)$, $(0, 1)$. The extra relators for H_n can be taken as T^{2^e}, U^{2^e} in the square case, and $T^{2^e}U^{2^e}, T^{2^e}U^{-2^e}$ in the diagonal case. After deleting any marker, both T, U have order at most two: deleting a leaves conjugates of c, deleting c leaves conjugates of a, and deleting b leaves ac, ca. Therefore every lattice relator survives every deletion. The even Coxeter relators also survive. Thus H_n has Boolean deletions and IP by UA.D.

Moreover,

$$\gamma_3(W) = 2\mathbb{Z}^2, \qquad \gamma_3(H_n) = 2\mathbb{Z}^2/\Lambda_n, \qquad |\gamma_3(H_n)| = 2^{n-5}. \tag{S2}$$

Indeed reduction modulo $2\mathbb{Z}^2$ makes the adjacent squares central; conversely $[[a, b], c]$ and $[[b, c], a]$ give the two doubled unit translations, up to signs. Thus H_5 is class two of order 32. The squared translations are not always independent cyclic subgroups: in H_6 they coincide and have order two. Formula (S2) uses a lattice index and does not require independence of those cyclic subgroups. □

Lemma UA.SquareRecognition (all finite two-group square markings). Every finite two-group string C-group of type $\{4, 4\}$ and order 2^n is marked-isomorphic to H_n; necessarily $n \geq 5$.

Proof. Its two adjacent dihedral parabolics have orders eight and intersect in order two, so their product already has 32 elements. Thus $n \geq 5$. The complete affine presentation of UA.Affine maps onto the given group G; write N for its kernel. IP gives irredundancy, and UA.Phi identifies G/G' with the elementary abelian group on its three markers. Summing those three binary coordinates gives a character sending each reflection to the nonidentity element of C_2. In the affine group this character is orientation parity. Hence every element of N preserves orientation.

The kernel contains no nonidentity torsion. A finite-order plane isometry has a fixed point, obtained for example by averaging the orbit of any point. The square reflection group moves that point into a closed fundamental right triangle. The stabilizer of a point in that triangle is trivial in the interior, generated by its wall reflection on an open edge, or the finite dihedral group generated by the two walls meeting at a vertex. These stabilizers are conjugates of the proper marked parabolics. Their maps into G are injective by its exact pair orders and IP, so none contains a nonidentity element of N.

An orientation-preserving plane isometry is a translation or a rotation. The linear part here has finite order in D_8, so a nontranslation rotation has finite order. Thus all elements of N are translations. Since N is normal of finite index, it is a D_8-invariant sublattice of $\mathbb{Z}^2$, of index 2^{n-3}. Under $\mathbb{Z}^2 = \mathbb{Z}[i]$, invariance under a quarter-turn makes it an ideal. The Euclidean algorithm in $\mathbb{Z}[i]$ makes it principal; its generator has norm a power of two. As 2 is a unit times $(1 + i)^2$, unique factorization shows that this generator is a unit times a power of $1 + i$. Even powers give the square lattice and odd powers the diagonal lattice in UA.Square. There is one such ideal at each two-power index. Therefore $N = \Lambda_n$ and the marked quotient is H_n. □

Proposition UA.SquareDecomposition (the two square decompositions). For every integer $q \geq 1$, let $M_2(q)$ and $M_1(q)$ be the groups obtained by adjoining respectively $(bcba)^q = 1$ and $(cba)^{2q} = 1$

to the complete presentation (S1). They are the square and diagonal lattice quotients, and have the decompositions (S3) and (S4) below. Here D_{2q} is the dihedral group of order $2q$, including $D_2 = C_2$ and $D_4 = C_2^2$.

Proof. Here bcba is translation by $(1, 0)$, and $(cba)^2$ is translation by $(1, 1)$. Their signed-permutation orbits therefore generate exactly the square and diagonal lattices in (S3)–(S4). This identifies those complete presentations with the indicated lattice quotients.

In the square case, $\langle a, c^b\rangle$ acts on the first coordinate and $\langle a^b, c\rangle$ on the second. Their translations have order q; each is D_{2q}, since its reflection has nonidentity linear part, also when q=1. They commute and intersect trivially. The reflection b interchanges the two factors and lies outside their product. They and b generate all three original generators. Hence

$$W/q\mathbb{Z}^2 \cong (D_{2q} \times D_{2q}) \rtimes C_2. \tag{S3}$$

In the diagonal case, put $A_0 = \langle b^a, b^c\rangle$ and $B_0 = \langle b, b^{ac}\rangle$. Their reflecting axes are respectively $x+y = 0, 1$ and $x-y = 0, -1$. The first factor has translations along $(1, 1)$, the second along $(1, -1)$, of order q in each case. Each reflection retains its nonidentity linear part, so the two factors have order 2q also at q=1. All cross-generators commute. Their linear parts show that an element of their intersection must be a translation; the independence of the two diagonal translations modulo the defining lattice then makes that intersection trivial. Thus $A_0B_0 \cong D_{2q} \times D_{2q}$.

Both a and c exchange the two factors by conjugation, so normalize their product. The two individual reflections a,c do not belong to the product by their linear parts. The only remaining possible nontrivial intersection with $\langle a, c\rangle$ is ac. Every translation part of an element of A_0B_0 with linear part -1 has coordinates of the same parity, whereas ac has translation part $(0, 1)$. The diagonal lattice preserves this parity condition, so ac is also outside the product. Finally b already lies in B_0, so the two factors and $\langle a, c\rangle$ generate the full group. Consequently

$$W/q\langle(1, 1), (1, -1)\rangle \cong (D_{2q} \times D_{2q}) \rtimes (C_2 \times C_2). \tag{S4}$$

These arguments specify the actions by the original affine generators and prove the decompositions, rather than only equality of orders. □

10 Changing the orders on selected pairs

Theorem UA.Edge (dihedral attachments). Let $B = \langle b_i : i \in I\rangle$ be marked by nonidentity involutions and have every Boolean deletion. Select a family $\mathcal{E}$ of distinct pairs $S = \{i, j\}$ for which the product order $q_S = |b_ib_j|$ is finite and even. For each selected pair choose a positive multiple m_S of q_S, and attach the marked dihedral quotient

$$D_{2m_S} \longrightarrow B_S = D_{2q_S}.$$

The full compatible product of UA.SK is generated by its prescribed marker tuples. It has the intersection property and all Boolean deletions. Every selected pair has product order m_S, and every unselected pair retains its original product order. If B is finite, then for every $J \subseteq I$,

$$|H_J| = |B_J| \prod_{\substack{S\in\mathcal{E}\\ S\subseteq J}} \frac{m_S}{q_S}. \tag{E1}$$

If the base marking gives a regular hypertope, so does the constructed marking, and its full type-preserving automorphism group is H.

Proof. A group generated by two nonidentity involutions whose product has finite order q_S is the dihedral group D_{2q_S}; the normal form alternates between powers of the product and a single reflection. The indicated quotient exists because $q_S \mid m_S$. It is injective on each proper local parabolic, which is

either trivial or generated by one reflection. UA.D supplies IP in the base, and all input deletions are part of the hypothesis. Thus UA.SK applies. The pair supports form an antichain, so UA.SK.Local identifies every selected pair subgroup with its full local dihedral group. UA.SK.Delete gives the asserted deletions. The kernel on a selected pair has order m_S/q_S, and (E1) is UA.SK's order formula.

For an unselected pair, each extra coordinate of its product is a reflection or the identity, because that coordinate's support cannot contain both markers. If the base product has finite order q, its image under a deletion keeping just one of the two markers has order two, so q is even. The extra coordinates therefore do not enlarge its order. If the base product has infinite order, its lift has infinite order by projection. This proves the assertion for every unselected pair.

All local markers are dihedral reflections, hence involutions. Under the final hypothesis, UA.SK.Geometry gives the geometric conclusion and identifies the full type-preserving automorphism group. □

11 Amalgams and primary torsion

Fix injective homomorphisms $i_A : C \to A$ and $i_B : C \to B$. Let $A * B$ be the free product and let

$$P = (A * B)/\langle\langle i_A(c) i_B(c)^{-1} : c \in C \rangle\rangle. \tag{A1}$$

We use the amalgam normal-form theorem in the following precise form: the canonical maps of A and B into P are injective; an alternating reduced word with factors outside the common copy of C has nonzero reduced length and is nonidentity; every element is conjugate either into a factor or to a cyclically reduced alternating word of length at least two, whose positive powers remain reduced with proportionally increasing length. This is a retained classical theorem.

Theorem UA.Amalgam (universal property and torsion). The group P in (A1), with its canonical maps j_A, j_B, is a pushout of $A \leftarrow C \rightarrow B$ in the category of groups: for every group D and homomorphisms $f_A : A \to D$, $f_B : B \to D$ satisfying $f_A i_A = f_B i_B$, there is a unique homomorphism $f : P \to D$ with $f j_A = f_A$ and $f j_B = f_B$.

Let p be a prime. Say that a group has p-primary finite torsion when every finite-order element has p-power order. This property passes to subgroups, to extensions with finite p-group quotient, and to amalgamated free products of groups having the property. If $i_A(C) \neq A$ and $i_B(C) \neq B$, then P is infinite. If A and B are finitely generated, then so is P.

Proof. The free-product universal property gives a unique homomorphism $A * B \to D$ extending f_A, f_B. Agreement on C puts every defining relator in its kernel, so it factors through P. The images of A and B generate P; hence this factorization is unique. The equality $j_A i_A = j_B i_B$ follows from the defining relations, completing the pushout property.

The torsion assertion for subgroups is immediate. For an extension $1 \to K \to E \to R \to 1$, with R a finite p-group and K having p-primary finite torsion, take any finite cyclic subgroup $\langle x \rangle$ of E. Its intersection with K is a cyclic p-group, and its image in R is a p-group. The order formula for this restricted exact sequence makes $|x|$ a power of p.

For an amalgam, the normal-form theorem places a torsion element, up to conjugacy, in a factor: the alternative cyclically reduced word has powers of unbounded reduced length and cannot have finite order. The factor hypothesis therefore proves the torsion assertion. If the common subgroup is proper in both factors, choose $a \in A \setminus i_A(C)$ and $b \in B \setminus i_B(C)$. The words $(ab)^n$ are reduced of length $2n$, so are all distinct; this proves infinitude. Finally, the images of finite generating sets of A and B generate their free product and its quotient P, proving finite generation. □

The simultaneous fiber product of UA.SK has its displayed compatibility equations and its proved generation theorem. It is not the pushout in UA.Amalgam. A graph of proof dependencies is neither of these group constructions.

12 Recognizing an affine presentation

We write an integral affine transformation as (M, t), acting by $v \mapsto Mv + t$, with multiplication

$$(M, t)(N, u) = (MN, t + Mu). \tag{1}$$

Let E be a group generated by finitely many such transformations, and let $Q \le \mathrm{GL}_d(\mathbb{Z})$ be its finite linear image. Its translation kernel L is a subgroup of $\mathbb{Z}^d$, invariant under Q. A translation is written $t_v = (I, v)$.

Lemma AL.1 (fixed Schreier lattice). Choose a word lift $\hat{q} \in E$ for each $q \in Q$, with $\hat{1} = 1$. Then L is generated by the translations

$$\widehat{\overline{sq}}^{-1} s\hat{q}, \qquad s \text{ a distinguished generator}, \quad q \in Q. \tag{2}$$

Proof. The products have trivial linear image. Write a partial word as $\hat{q}n$, with $n \in L$. Left multiplication by s changes this to $\widehat{\overline{sq}}tn$, where t is the transition in (2). Induction expresses a word of trivial linear image as a product of these transitions; its final representative is $\hat{1} = 1$. Inverses introduce no extra generator: the transition for s^{-1} at q is the inverse of the transition for s at $\overline{s}^{-1}q$. Thus the induction applies to words with positive and negative letters. Every transition itself is in L, so they generate exactly L. □

The normalization in (2) matters. The raw differences $s\hat{q}\widehat{\overline{sq}}^{-1}$ need not generate L without conjugation. Each is a conjugate of the corresponding transition by $\widehat{\overline{sq}}$. Consequently their Q-orbits generate L. In particular, if a proposed Q-invariant lattice contains every raw difference, it contains every normalized transition and hence L. If exhibited raw differences also span the proposed lattice, equality follows. This invariant-lattice version justifies the raw-difference format of the unchanged earlier $\{6, 6\}$ checker [C66]. The new $\{6, 4\}$ checker uses (2) directly.

This is a finite proof of a lattice identity. Containment in a proposed lattice is checked on the normalized transitions, or on the raw differences with the stated invariance condition. Equality follows when independent integer combinations give a basis of that lattice, or when the greatest common divisor of the full-rank minors gives its exact index. Sampling finite quotients is not a replacement for this step.

Lemma AL.2 (presentation bridge). Let U be a finitely presented marked group whose relators hold on the affine generators of E. Suppose their linear images define the same epimorphism to Q. If an exact Reidemeister–Schreier presentation proves $\ker(U \to Q) \cong \mathbb{Z}^d$, and the affine translation kernel L has rank d, then the marked epimorphism $U \to E$ is an isomorphism.

Proof. The restriction maps $\ker(U \to Q)$ onto L: an affine translation has a preimage in U, and its linear image is trivial. This is a surjection between free abelian groups of rank d. Tensoring with $\mathbb{Q}$ shows that its kernel has rank zero. A subgroup of a free abelian group is torsion-free, so that kernel is trivial. The remaining kernel of $U \to E$ lies in this subgroup, and is therefore trivial as well. □

The hypothesis concerns the actual kernel presentation. Computing only its abelianization would not establish it: a nonabelian group can have free abelian abelianization of rank d. Our applications retain the exact rewriting and Tietze algorithms used to verify this finite presentation step. It is a computer-assisted input, not a Lean proof.

13 All parabolic intersections in an affine period quotient

The finite adjacent subgroups in the polyhedral applications have faithful linear images. An arbitrary parabolic need not have that property: it can contain translations. We therefore state the intersection calculation before making the faithful-image specialization. It uses the actual affine group, whose identification with a presented group is the separate obligation in AL.2.

Let $E = \langle r_i : i \in I \rangle \leq \mathbb{Z}^d \rtimes Q$, where I and the linear image Q are finite. Multiplication is $(q,u)(q',v) = (qq', u + qv)$. Write $L \leq \mathbb{Z}^d$ for the translation kernel of E. For $J \subseteq I$ put

$$E_J = \langle r_i : i \in J \rangle, \qquad Q_J = \mathrm{im}(E_J \to Q), \qquad K_J = \{v : (1,v) \in E_J\} \leq L.$$

Choose, for each $q \in Q_J$, an element $(q, u_J(q)) \in E_J$, with $u_J(1) = 0$. These are finite choices of affine word lifts, not assumptions about a splitting of E_J. The entire fiber over q is

$$\{(q,u) : u \in u_J(q) + K_J\}. \tag{5}$$

Indeed the quotient of two elements with the same linear part is their translation difference; conversely left multiplication by $(1,k)$ adds any $k \in K_J$ to the translation part.

Let $M \leq L$ be Q-invariant. It is then a normal translation subgroup of E. Write $\overline{E} = E/M$ and $\overline{E}_J$ for the image of E_J. The subgroup M can have any rank, including zero.

Theorem 1 (AA.ParabolicPeriods)**.** *The marking of E/M has the intersection property if and only if, for every $J, T \subseteq I$, with $C = J \cap T$, both conditions hold:*

$$(K_J + M) \cap (K_T + M) = K_C + M, \tag{6}$$

$$u_J(q) - u_T(q) \notin K_J + K_T + M \quad (q \in (Q_J \cap Q_T) \setminus Q_C). \tag{7}$$

More generally, a common linear part $q \in Q_J \cap Q_T$ occurs in $\overline{E}_J \cap \overline{E}_T$ exactly when

$$u_J(q) - u_T(q) \in K_J + K_T + M.$$

When it occurs, its translation fiber is a coset of $((K_J + M) \cap (K_T + M))/M$.

Proof. Modulo M, equation (5) becomes the coset $(u_J(q) + K_J + M)/M$. Equality in E/M preserves the linear part, since M consists of translations. Two cosets of additive subgroups intersect if and only if the difference of their representatives belongs to the sum of those subgroups. When nonempty, their intersection is a coset of the intersection of the subgroups. This proves the two fiber assertions.

For $q \in Q_C$, a lift $(q, u_C(q)) \in E_C$ belongs to both E_J and E_T. It can therefore be used as a common representative of their two translation cosets. The intersection fiber is

$$\big(u_C(q) + (K_J + M) \cap (K_T + M)\big)/M,$$

whereas the fiber of $\overline{E}_C$ is $(u_C(q) + K_C + M)/M$. Condition (6) makes them equal. Condition (7) excludes every other possible linear part. Together these prove $\overline{E}_J \cap \overline{E}_T = \overline{E}_C$.

Conversely that equality, at the identity linear part, implies (6): the relevant additive groups all contain M, so equality of their quotients by M is equality of the groups themselves. For a linear part outside Q_C its intersection fiber must be empty, which gives (7). Applying this to every pair of index sets proves the equivalence. Changing any chosen lift adds an element of K_J or K_T and does not change the membership test. □

All data in this criterion can be obtained by finite exact calculations for a fixed period lattice. Enumerate Q_J from the linear generator matrices, retain a word lift of each element, and apply AL.1 to E_J and Q_J. Its normalized Schreier translations generate exactly K_J. Integer lattice sum, intersection and membership calculations then decide (6)–(7). No finiteness of E_J and no injectivity of $E_J \to Q_J$ is assumed. For a parameterized family M_m the same equations remain exact, but an all-parameter solution of them still needs a proof; finitely many successful periods do not establish it.

The first condition cannot be omitted when parabolics contain translations. Take $E = \mathbb{Z}^2$ with markers e_1, e_2 and trivial Q. For $M = \langle e_1 - e_2, 2e_2 \rangle$, the two distinct rank-one parabolics have the same nontrivial image in $E/M \cong C_2$, so IP fails. Here $K_{\{1\}} + M = K_{\{2\}} + M = \mathbb{Z}^2$, whereas $K_\emptyset + M = M$. There is no nonidentity linear part to detect this failure. By contrast, $M = 2\mathbb{Z}^2$ gives the two distinct

coordinate factors of C_2^2 and satisfies the criterion. These examples concern general marked groups; the theorem does not require involutory markers.

This statement extends the previously proved rank-three period argument to all marked parabolics. Its proof is the classical intersection formula for cosets of additive subgroups. The extension and the explicit hypothesis separation are part of this edition; no claim of historical priority is made. Existing formal period certificates cover the previously stated faithful-parabolic monotonicity, not this full arbitrary-parabolic theorem.

14 Periods and intersections of distinguished subgroups

Let $w_1, \dots, w_d \in U = E$ be translation words with linearly independent vectors $v_1, \dots, v_d$. Write $\Lambda = \sum_i \mathbb{Z} v_i \le L$ and suppose it is Q-invariant. For a positive integer m, adjoining $w_i^m = 1$ to the complete presentation of U gives exactly

$$E_m = E/(m\Lambda). \tag{3}$$

Indeed, the normal closure of the powers is $m\Lambda$: conjugation by an affine element acts on translations by its linear part, and this lattice is already invariant. Hence

$$|E_m| = |Q|[L : \Lambda]m^d. \tag{4}$$

This proves equality for the presented group, not merely the order of a chosen quotient. If Q is solvable, every E_m is solvable.

Suppose now that the marking has rank three with distinct nonidentity involutions, its endpoints commute, both adjacent products have order at least three, and its two adjacent subgroups $A = E_{01}$ and $B = E_{12}$ are finite and inject into Q. For each $q \in \overline{A} \cap \overline{B}$, let $(q, t_A(q))$ and $(q, t_B(q))$ be their unique affine lifts. Put

$$\Delta(q) = t_A(q) - t_B(q). \tag{5}$$

Theorem AL.3 (period-lattice intersection criterion). The images of A and B in E_m meet precisely in the elements indexed by

$$\{q \in \overline{A} \cap \overline{B} : \Delta(q) \in m\Lambda\}. \tag{6}$$

In particular, if $A \cap B = \langle r_1 \rangle$ and every other value $\Delta(q)$ lies outside Λ, then every E_m, $m \ge 1$, has the full rank-three intersection property and the same adjacent orders as E. Its order is (4).

Proof. Apply the fiber assertion of AA.ParabolicPeriods to $J = \{0, 1\}$, $T = \{1, 2\}$ and $M = m\Lambda$. Faithful linear images give $K_J = K_T = 0$, so its general coset-membership condition is exactly (6). They also show that both adjacent subgroups remain injective in the quotient. Since $m\Lambda \subseteq \Lambda$, the stated exclusions leave only the common involution. The rank-three test gives all subset intersections. Adjacent orders and (4) follow from the injectivity and the full translation index already established. □

The proof separates two possibilities often hidden by equal group-order counts. The point-group intersection may already have order two, as in the $\{6, 4\}$ application. Or it may be larger, with a short list of nonzero translation differences, as in the $\{6, 6\}$ application. Those differences must be excluded from the period lattice explicitly.

One coarse quotient controls all smaller periods

The same argument has a group-theoretic formulation which is useful independently of coordinates. Let A, B, L be subgroups of a group E, with

$$L \cap A = L \cap B = 1.$$

Let $N \trianglelefteq E$ lie in L. If the images of A, B in E/N intersect exactly in the image of $A \cap B$, then the same statement holds in E/M for every normal subgroup $M \leq N$.

Indeed, suppose $a \in A$, $b \in B$ have the same image modulo M. They have the same image modulo N, so there is $c \in A \cap B$ with $aN = cN = bN$. Then $ac^{-1} \in N \cap A = 1$ and $bc^{-1} \in N \cap B = 1$, proving $a = b = c$. The reverse inclusion is immediate. In the affine setting L is the translation kernel, $N = \Lambda$ and $M = m\Lambda$. Point-faithfulness supplies precisely the two trivial intersections. This statement, for an arbitrary indexed family of smaller normal periods, is verified in `AffinePeriodIntersections.lean`. The finite coordinate checks which identify L, Λ and the initial overlap remain separate mathematical obligations.

15 Changing one adjacent order

Let $(B; a, b, c)$ be a rank-three string C-group of type $\{h, q\}$, with h, q even, and suppose the map

$$\rho : B \longrightarrow B_{12}, \qquad a \mapsto 1, b \mapsto b, c \mapsto c \tag{7}$$

is a retraction. For k a positive multiple of q, take the standard marked epimorphism $D_{2k} \to B_{12} = D_{2q}$. The compatible product

$$F = \{(g, d) \in B \times D_{2k} : \rho(g) = \overline{d}\} \tag{8}$$

has markers $(a, 1), (b, \tilde{b}), (c, \tilde{c})$. The evenness of h is forced by (7); it also ensures that the first pair has unchanged order h.

Theorem AL.4 (one-edge extension). These markers generate F. If B is finite, its order is $|B|k/q$. Its type is $\{h, k\}$, and it has the intersection property. The projection onto B is injective on the first adjacent subgroup. If B is solvable, so is F.

Proof. Apply UA.SK to the single support $S = \{1, 2\}$, using the given deletion and the marked dihedral epimorphism $D_{2k} \to D_{2q}$. Its restrictions to both reflection subgroups are injective, which verifies every proper local subgroup. Its kernel has order k/q. The theorem gives generation of the whole compatible product, its order and IP. UA.SK.Local identifies the second adjacent subgroup with D_{2k}; the first has trivial projection kernel by (K2), since it does not contain S. Thus its order and adjacent product order remain unchanged. Explicitly the other local coordinate of that product has order two, which divides the even integer h. The cyclic projection kernel and the solvability of the base prove solvability. □

Theorem AL.4 is a specialization of UA.SK; it is included to make clear that only the indicated deletion is needed here. More simultaneous edges use the supported coordinate kernels of that theorem.

There is a useful exact presentation variant. Suppose a complete marked presentation for B contains $(bc)^q = 1$. Replace that relation by $(bc)^k = 1$, retain the remaining relators, and add $[a, (bc)^q] = 1$. Call the resulting presented group H. If every retained relator holds in (8), then $H = F$ as a marked group. Here the local dihedral presentation holds in H by the involution relations and $(bc)^k = 1$. Its kernel word $(bc)^q$ generates a normal subgroup: b, c invert it and the new relation makes a centralize it. Killing that word gives exactly the original marked presentation of B. These are all four conditions of UA.Recognition, so $H = F$ even when the base is infinite. No finite order comparison is needed.

16 Recognizing supported extensions of affine quotients

The following interface combines the two recognition steps. It separates a presentation of the affine base from the presentation of a group with larger local sections. The latter is not identified merely by displaying a finite image of the right type.

Theorem 2 (AA.PeriodExtension)**.** *Let I be finite and let $U = \langle r_i : i \in I \rangle$ be a finitely presented marked group with an affine marked image $E \le \mathbb{Z}^d \rtimes Q$, where the finite linear images agree. Suppose the actual kernel of $U \to Q$ has a presentation identifying it with $\mathbb{Z}^d$, and that the translation kernel L of E has rank d. Let $M \le L$ be Q-invariant, with an explicitly given finite lattice basis represented by words in U.*

Suppose the equations of AA.ParabolicPeriods hold for M. Put $B = E/M$. Choose a finite antichain $\mathcal{F}$ of label supports. For each $S \in \mathcal{F}$, suppose there is a deletion $p_S : B \to B_S$ and a marked epimorphism $q_S : V_S \to B_S$ from a group generated by the markers on S, injective on every proper marked subgroup of V_S. Write $N_S = \ker q_S$.

Let P be a marked presentation with the following properties:

1. *Its relators hold in the compatible product of B and the V_S with respect to p_S, q_S.*
2. *Every local presentation for V_S holds in P.*
3. *The image in P of each N_S is normal in P.*
4. *Killing these local kernel images leaves exactly the presentation of U with the basis words of M set equal to one.*

Then P is marked-isomorphic to that full compatible product. It has IP, and its parabolic projection kernels are the coordinate products

$$\ker(P_J \to B_J) = \prod_{S \subseteq J} N_S.$$

Each local subgroup P_S is exactly V_S. If $[L : M]$ and all N_S are finite, then

$$|P| = |Q|[L : M] \prod_{S \in \mathcal{F}} |N_S|, \qquad |P_J| = |B_J| \prod_{S \subseteq J} |N_S|.$$

The geometric and deletion conclusions hold precisely under the additional hypotheses stated in UA.SK.Geometry and UA.SK.Delete, respectively.

Proof. AL.2 identifies U with E using the actual free abelian kernel presentation and its surjection onto the rank-d translation lattice. The normal closure of the chosen basis translations of M in E is M: it contains their additive span, and invariance under the linear image shows that this span is already normal. Thus adjoining these words identifies the whole base presentation with B, not only with a quotient of B. AA.ParabolicPeriods gives IP in B.

The stipulated deletions and proper-parabolic injectivity are exactly the hypotheses of UA.SK. That theorem constructs and generates the full compatible product, identifies every projection kernel, and proves IP. The four presentation hypotheses are exactly UA.Recognition. Its proof uses normality and the faithful separate local coordinates to exclude every remaining kernel; it proves P is that compatible product. The antichain clause UA.SK.Local identifies its local subgroups. When the displayed indices are finite, $|B| = |Q|[L : M]$, and UA.SK gives the stated orders. The last sentence invokes the additional proved hypotheses of the two named corollaries; IP alone does not establish a hypertope's flag property. □

The theorem also applies with no local supports, in which case $P = B$. For an affine base with an elementary normal-form proof, that proof can replace the Schreier computation in the first paragraph: the required input is the exact marked isomorphism $U = E$, not a particular algorithm for proving it. This is how the square groups enter the two-power applications. The cubic $\{6, 4\}$ and $\{6, 6\}$ bases instead retain their two full exact kernel-presentation programs. Their lattice and overlap programs verify the separate arithmetic hypotheses. No one of these certificates replaces the others.

This theorem assembles already proved normal-kernel and supported-extension arguments with the period criterion. The eight applications below retain their source authorship. The high-type and order-1024 classifications also need the forest, cyclic-core and exhaustive-kernel proofs; they are not consequences of this construction theorem alone.

17 Retained inputs, applications and attribution

The proofs use the elementary subgroup, quotient and isomorphism theorems. Square recognition also uses the Euclidean algorithm and unique factorization in $\mathbb{Z}[i]$, and the elementary point stabilizers in the square reflection tiling described in its proof. The further classical inputs are explicitly limited: Burnside's basis theorem for finite p-groups; their nilpotence; generation of the lower-central graded Lie ring by degree one and its multilinear commutator rules; and the precise amalgam normal-form theorem stated in the amalgam section. The Frattini, forest, finite-quotient and torsion deductions are proved in full above. No finite group classification, atlas census or numerical search result supplies a premise to these theorems.

For applications to abstract regular polytopes we retain the following standard correspondence [MS02, Section 2E]. A group generated by an ordered list of nonidentity involutions, with the string commutation rules and IP, is the full automorphism group of a regular abstract polytope with that distinguished marking. Its rank is the length of the list, its type consists of the adjacent product orders, and its number of flags is the group order. A generator-preserving epimorphism between such groups induces a rank- and adjacency-preserving covering of their associated polytopes. These are geometric input theorems, not new conclusions of the present paper. By contrast, the coset-geometric recognition and coordinate-kernel Helly statements needed for the hypertope branch are proved in the incidence section with their full hypotheses.

The supported-extension construction supplies a common proof interface for the two-power constructions studied in [HFL20, HFL19a, HFL19b, CFHS25]. The presentation-recognition theorem supplies the additional normality and exact-base conditions needed to identify a compatible product with a complete universal presentation; local construction alone does not do this. The forest quotient gives the canonical group underlying the main quotient and cover conclusions of [HS26], and the finite-kernel parameter space gives the exhaustive part of the order-1024 reconstruction associated with [GLP18]. Their exact source statements, literal corrections, numerical certificates and named-object identifications belong to the separate applications.

The supported-kernel induction, forest quotient and affine models are reused complete proofs. The normal-kernel recognition theorem replaces repeated finite-order comparisons, and the mix criterion separates joint coordinate detection from generation of a fiber product. Every application still proves its local defining relations, normality and base identifications. Original source results and borrowed geometric theorems keep their attribution.

Existing Lean certificates cover declared components of the kernel, supported-extension and Helly arguments. They do not presently formalize this whole paper: in particular, the full forest quotient proof, the affine model and the geometric recognition/application interfaces are not all connected in Lean. Their written proofs above remain part of the mathematical obligation. The accompanying certificate scope distinguishes inherited checked declarations from any freshly checked application components.

The mixed-prime application also retains Burnside's $p^a q^b$ solvability theorem, the exact Lucchini cyclic-core bound as quoted with its hypotheses, and the complete SmallGroups catalogue at order 384. The affine bridge uses actual Reidemeister–Schreier and Tietze computations of two kernels, not their abelianizations. The graph-symmetry application retains Frucht's theorem and the complete SmallGroups catalogues at orders 40,48,56 for its isolated rank-three minimum. These are application inputs; none is silently used to prove the general lattice-coset or normal-kernel theorems.

All eleven complete required program/data files are retained once in this volume: four files for the two-group classification (including its complete named-presentation data and runner), six for the mixed-prime affine families, and one for the graph-symmetry minimum. The three complete appendices print those bytes. Operational drivers and all previously scoped Lean files are separately inventoried. Existing successful semantic and computational reviews are reused for unchanged bytes; this consolidation does not claim to have reexecuted every old census or to have formally verified the new arbitrary-parabolic lattice criterion.

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Rotation certificates and affine period families

A shared supplement to the complete marked-kernel theory

Private draft; independent scientific review pending

The complete common paper [3] remains an acquired premise, including AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry. This supplement adds a finite rotation certificate and moves the complete previously accepted cubic, square and finite presentation proofs here so that both [1] and [2] use a common proof. No application is an uncharged premise of another. The exact migration map preserves every moved paragraph and its whole program and data dependencies. The constructions and prior principal results remain attributed to their source authors. The finite certificates are required computations, whereas sample calculations never establish an unbounded parameter assertion.

1 Exact rotation presentations and coset certificates

The following certificates isolate the proof shared by the finite regular and chiral rows. It does not identify a finite image with a presented group, and it does not infer flag transitivity from subgroup intersections alone. Every source-specific table row remains a separate certificate obligation.

Proposition 1 (Exact regular completion)**.** *Let $r \geq 3$, let $P = \langle a_1, \dots, a_{r-1} \rangle$ be a finite group with its specified marking, and put $a_0 = 1$. Suppose that simultaneous inversion of the a_i extends to an automorphism σ of P. Form $H = P \rtimes \langle t \rangle$, with $t^2 = 1$ and $ta_it = a_i^{-1}$, and put $\rho_i = ta_i$ for $0 \leq i < r$. Suppose that these involutions have the full marked subgroup intersection property and that their maximal parabolics $H_i = \langle \rho_j : j \neq i \rangle$ have typed coset Helly. Then the standard H-coset geometry is a regular hypertope with full type-preserving group H, of order $2|P|$. It is isomorphic to the source rotation coset system with parabolics*

$$P_i = \langle a_j^{-1} a_k : j, k \neq i \rangle.$$

Thus P has exactly two orbits on its chambers, and H is its regular completion. For $i \geq 1$ the displayed P_i equals $\langle a_j : j \neq i \rangle$, and $P_0 = \langle a_1^{-1} a_j : j \geq 2 \rangle$.

Proof. The ρ_i are involutions and generate H, since $\rho_0 = t$ and $\rho_0 \rho_i = a_i$. The sign homomorphism $H \to C_2$ has kernel P. Its restriction to H_i is onto, and its kernel is generated by the even words $\rho_j \rho_k = a_j^{-1} a_k$. Hence $H_i \cap P = P_i$ and $[H_i : P_i] = 2$. Every right coset of H_i has an even representative, so $P_i g \mapsto H_i g$ is a bijection on elements of each type. It also preserves incidence. One implication is immediate. Conversely, if two H-cosets have a common odd element h, choose a label k different from their two types. Then $\rho_k h$ is an even common element. This uses $r \geq 3$ and proves the claimed incidence isomorphism without assuming equality of arbitrary products or quotient parabolics.

Full marked IP and typed coset Helly give the regular hypertope by the common chamber construction: Helly gives flag completion and flag transitivity; IP gives trivial chamber stabilizer and two elements in every panel; the generating property in each marked subgroup gives residue connectivity. A type-preserving automorphism fixing a chamber fixes each adjacent chamber by thinness, and then all chambers by gallery connectivity. Consequently the displayed chamber-regular H is the full type-preserving automorphism group. Its index-two subgroup P has exactly

two chamber orbits. The formulas for P_i follow by including $a_0 = 1$ when $i \neq 0$, and by taking differences from a_1 when $i = 0$. □

An exact presentation certificate must first identify P itself. In the finite rows, a completed coset enumeration or an exact fp-group isomorphism supplies an upper bound; an actual permutation image supplies the matching lower bound. A numerical image of the expected order by itself supplies only the lower bound. The regularity test then checks all marked subgroup intersections and the separate normalized coset-Helly identities in H. Neither a group order nor a source structural abbreviation replaces these checks.

For a chiral row the last proposition is not applicable. We must instead certify the rotation coset geometry itself, including thinness, residue connectivity, two chamber orbits with adjacent chambers in opposite orbits, and the absence of simultaneous inversion. Its full automorphism order is then $|P|$, whereas a regular row has full order $2|P|$. The source tables and the literal rotation presentations must retain this distinction.

A bounded certificate for chiral rows

Here all cosets are right cosets. For a finite group P with its three rotation generators, use the four subgroups P_i above and write $K_J = \bigcap_{j\in J} P_j$. For $i \notin J$, choose $j_0 \in J$ and a right transversal T of $P_i \cap P_{j_0}$ in P_{j_0}. Set

$$c(i, J) = \#\{g \in T : g \in P_i P_j \text{ for every } j \in J\}.$$

This counts exactly the type-i elements incident with the base flag of types J, provided that base flag has been fixed; the representatives lie in one finite proper parabolic, not in the entire ambient group.

Proposition 2 (Two chamber orbits, with independent connectivity checks)**.** *Suppose the marked pair-difference subgroups have the full C^+ intersection property. Suppose additionally that*

1. *$c(i, J) = [K_J : K_J \cap P_i]$ for every $|J| = 2$, and $c(i, J) = 2$ for every $|J| = 3$;*
2. *P is generated by the four P_i, and each P_f is generated by its subgroups $P_f \cap P_j$, $j \neq f$;*
3. *for each pair of types F, the graph with two vertex sets indexed by K_F, one for each remaining type i, j, and edges*

$$h \sim k \quad \Longleftrightarrow \quad hk^{-1} \in P_i P_j \qquad (h, k \in K_F)$$

is connected.

Then the rotation coset system is a hypertope. The group P acts freely with two chamber orbits, and adjacent chambers lie in opposite orbits. It is chiral with full type-preserving group P if simultaneous inversion of the three marked generators does not extend to an automorphism of P.

Proof. Pair flags are transitive in any coset system. The $|J| = 2$ equalities say that all eligible cosets are supplied by K_J, so all three-type flags are transitive as well. The C^+ intersection identities give trivial three-type stabilizers: they are the pair-difference group on the one remaining label. The $|J| = 3$ equalities therefore give two completions of each panel, with trivial chamber stabilizer. Every shorter flag extends successively to a chamber, so this is a thin geometry with exactly two P-orbits on chambers. Two adjacent chambers cannot lie in the same orbit, since an element mapping one to the other would fix their panel and hence would be the identity.

The incidence graph of a coset system is connected when its defining subgroups generate its ambient group. Applied first to P, and then to P_f with its subgroups $P_f \cap P_j$, the second condition proves connectivity of the whole geometry and of every rank-three residue. The latter residue identification uses transitivity on three-type flags, so incidence of two elements through the fixed type-f element is represented by an intersection inside P_f. For a rank-two residue, the same transitivity represents each of its two vertex types by K_F; three-type stabilizers are trivial, so these labels are unique. Its actual incidence is exactly the graph in the third condition. This proves residual connectivity independently of the intersection checks.

Thinness and chamber connectivity make every chamber-fixing automorphism trivial. Thus the full type-preserving automorphism group has order at most $2|P|$. If it properly contains P, the latter has index two and is normal. Composing an extra automorphism with an element of P gives an involution t sending the base chamber C to its type-zero neighbor C^0: its square fixes C. For the source coset conventions,

$$C^0 = (P_0 a_1^{-1}, P_1, P_2, P_3).$$

Indeed $P_0 a_i^{-1} = P_0 a_1^{-1}$ for every $i > 0$; these cosets meet all three other base elements, and C^+ intersection excludes $a_1 \in P_0$. Right multiplication gives $C^0 a_i = C^i$, whence $C a_i^{-1} = (C^0)^i$. As type-preserving automorphisms commute with adjacency, $t a_i t$ and a_i^{-1} have the same effect on C and are equal. Thus an extra automorphism would induce simultaneous inversion on P. Its failure proves that P is the full group and the geometry is chiral. □

Every test above is finite and exact. The rank-two connectivity tests use only two copies of an intersection of proper parabolics. A bounded count is not extrapolated to untested presentations. For each literal finite presentation, simultaneous inversion can be decided by evaluating its full relator list on the inverse marked images: if all relations hold, the assignment defines an endomorphism whose square fixes every generator, hence an automorphism; a failed relation excludes it.

2 Literal local periods and actual marked quotients

A period appearing among global relators need not be the least period of the corresponding parabolic. This distinction is checked explicitly. For a finite local group generated by u, v, take one of the three standard rotation presentations

$$\begin{aligned} T_{63}(a,b) &= \langle u, v \mid u^6, v^3, (u^{-1}v)^2, (v^{-1}u^{-2})^a (vu^2)^b \rangle, \\ T_{36}(a,b) &= \langle u, v \mid u^3, v^6, (u^{-1}v)^2, (u^{-1}v^{-2})^a (uv^2)^b \rangle, \\ T_{333}(a,b) &= \langle u, v \mid u^3, v^3, (u^{-1}v)^3, (uv^{-1}u)^a (uv)^b \rangle. \end{aligned}$$

These retain the exact source parameter conventions. Their standard orders are $6(a^2+ab+b^2)$, $6(a^2+ab+b^2)$ and $3(a^2+ab+b^2)$. The usual faithful Euclidean triangle realization is retained here; for each finite row the program also computes the abstract local order independently and tests the exact marked epimorphism into the parabolic. Consequently no local isomorphism is inferred from an unmarked order match.

Let U, V denote the two translation words in the last relator of the chosen presentation. The program tests that their actual images commute, generate a normal subgroup L, and have index six (or three) in the local group. It determines $m = |U|$ and the least $b > 0$ for which $U^a V^b = 1$ with $0 \le a < m$. Then

$$\ker\big(\mathbb{Z}^2 \longrightarrow L,\ (i,j) \longmapsto U^i V^j\big) = \mathbb{Z}(m,0) + \mathbb{Z}(a,b).$$

Indeed reduce the second coordinate modulo b using (a,b) and then the first modulo m. Minimality leaves exactly mb different images; the program independently verifies $mb = |L|$. This

proves the entire lattice kernel, not merely two relations. The order of the complete abstract local presentation after adjoining U^m and U^aV^b is then computed and compared to the actual parabolic order. The generator map is onto and the equal orders prove an exact marked isomorphism.

For spherical rank-three residues the same method uses the complete triangle presentation $\langle u, v \mid u^p, v^q, (u^{-1}v)^r\rangle$, with $1/p + 1/q + 1/r > 1$, and an explicit onto generator map. Its exact order is computed. Every local result therefore specifies its marking and not just a name or a number. The full coset tests in the preceding section independently ensure that these are the actual residues of the global geometry.

When a printed word does not impose its claimed torus period, it is kept literally in the global presentation. The certificate computes the actual local lattice from the three triangle relations and the actual finite parabolic. It never inserts an intended torus word silently. Such a case is recorded separately from a proper collapse of an otherwise valid local period presentation.

3 Rotation presentations and the three shared marking adapters

Let W have involutory Coxeter generators $r_0, \dots, r_3$ and pair orders m_{ij}. Its sign map sends every generator to the nonidentity element of C_2. Reidemeister–Schreier rewriting with transversal $1, r_0$ gives the exact presentation of its even subgroup on $a_i = r_0 r_i$:

$$a_i^{m_{0i}} = 1, \qquad (a_i^{-1}a_j)^{m_{ij}} = 1.$$

The other Schreier generators are their inverses; rewriting both cosets of each Coxeter relator gives conjugates or inverses of these same relators. This proves a presentation identity, not a finite-image argument.

Suppose additional local rotation relators impose regular torus periods $(q, 0)$ or (q, q). Their normal closure inside each local triangle rotation group is invariant under its reflections: in the retained Euclidean lattice realization these are exactly the reflection-invariant period lattices. Their normal closure N in W^+ is then invariant under every global odd generator as well. For a local period and an odd local reflection r_j, write $r_0 = (r_0 r_j) r_j$; the first factor is even and the second preserves the local period subgroup. Thus $N \lhd W$, and

$$(W/N)^+ = W^+/N.$$

Consequently the displayed source rotation presentation, with these local period words, is exactly the even subgroup of the corresponding full reflection presentation. No additional, unverified reflected relation is being discarded.

This supplies the following explicit marking adapters for the shared families. All letters on the right are the involutions in the complete proof sections below; their specified order is preserved.

Family	x	y	z
Cubic (a, b, d, c)	ba	bc	bd
Square (a, b, c, d)	ad	ab	ac
Finite (a, b, c, d)	ca	cd	cb

For the cubic family the two period words become $c(bd)^2b$ and $a(bd)^2b$, with exponents two and s. For the square family, its four toroidal relations become, respectively,

$$(b(ad)^2a)^2, \quad (c(da)^2)^2, \quad (d(cb)^2)^{2s}, \quad (a(bc)^2b)^2.$$

For the finite family they become

$$(d(ca)^2c)^2, \quad (b(ac)^2a)^2, \quad (a(bd)^2)^2, \quad (c(db)^2d)^{3t} \quad (t = 1, 2).$$

These identities follow by cancellation of adjacent involutions, using the indicated commuting pair relations. In the square (s, s) relation, the two translation words commute, so their product raised to s equals $(d(cb)^2)^{2s}$. The parameter t here is exactly the shared finite theorem's parameter. Thus both universal presentations and the ordered local markings, not just their orders, transfer from the common proofs. Their subgroup intersections and flag geometry are unchanged under these explicit marker permutations.

4 The cubic family and its period lattice

Write $a = \rho_0, b = \rho_1, d = \rho_2, c = \rho_3$. Let E be generated by

$$a(x) = (x_2, x_1, x_3), \qquad b(x) = (x_1, x_3, x_2),$$
$$c(x) = (-x_2, -x_1, x_3), \qquad d(x) = (-x_2, -x_1, 1 - x_3).$$

Products act from right to left. Let Q be the signed permutation group in three coordinates, $\epsilon(q) \in \{0, 1\}$ the parity of its number of sign changes, and put $D = \{v \in \mathbb{Z}^3 : \sum v_i \equiv 0 \pmod 2\}$, $A = \{v \in \mathbb{Z}^3 : \sum v_i = 0\}$.

Lemma 1. *The group E is precisely $\{(q, v) \in Q \ltimes \mathbb{Z}^3 : \sum v_i \equiv \epsilon(q) \pmod 2\}$, with translation kernel D. Its marked presentation is*

$$\langle a, b, d, c \mid a^2, b^2, d^2, c^2, (ab)^3, (ad)^2, (ac)^2, (bd)^6, (bc)^3, (dc)^2, (c(bd)^2b)^2 \rangle.$$

Adjoining $(a(bd)^2b)^s = 1$ gives exactly $E_s = E/sD$, of order $48s^3$.

Proof. The linear maps generate all 48 signed permutations. Each affine map satisfies the parity condition, which multiplication preserves. Direct multiplication gives $u = a(bd)^2b = (1, (1, -1, 0))$; its Q-conjugates are the twelve roots $\pm e_i \pm e_j$, generating D. Hence the translation kernel is D. Surjectivity to Q proves the description of E. All eleven presentation relators hold.

Identification of the abstract presentation is separate. The complete program `cubic/kernel_certificate.g` forms its epimorphism onto the specified 48-element group and computes the actual kernel presentation by Reidemeister–Schreier rewriting and exact presentation simplification. It has three generators f_1, f_2, f_3 and exactly

$$[f_2, f_3], \qquad f_3 f_1^{-1} f_3^{-1} f_1, \qquad [f_1, f_2]$$

as relators. The full preimage words are emitted and independently evaluated by `cubic/certificate.py`. Their translations are $(0, 1, 1), (1, 0, -1), (-1, 0, -1)$, of determinant of absolute value 2. Thus this is the whole free abelian kernel, not its abelianization, and its image is D. AL.2 identifies the presented group with E. The exact GAP rewriting and simplification are declared software trust. Conjugating u^s generates exactly the already normal lattice sD, proving the quotient statement and its order. □

5 All parabolics and uniform intersections

Write $E_J = \langle \rho_i : i \in J \rangle$. A mask m denotes the set of positions of its nonzero binary digits. The only infinite proper parabolic is

$$E_{012} = \{(q, v) : q \in S_3 \times \{\pm I\},\ \sum v_i = \epsilon(q)\}, \tag{1}$$

where ϵ is zero on permutations and one on their negatives. Indeed each of a, b, d satisfies this formula, their linear image has order 12, and the conjugates of u by a, b generate A. This proves equality of every fiber. In particular $A \cap sD = sA$.

All other proper parabolics inject into their linear images. Complete closure of the actual affine transformations gives

m	0	1	2	3	4	5	6	8	9	10	11	12	13	14
$\lvert E_J\rvert$	1	2	2	6	2	4	12	2	4	6	24	4	8	48.

The maximal finite groups fix respectively $p = (-1/2, 1/2, 1/2)$ for E_{123}, $p_1 = (0, 0, 1/2)$ for E_{023}, and 0 for E_{013}. The first has linear image Q, and the last is the even-sign subgroup S_4.

Proposition 3. *For every $s \geq 3$ all 256 marked subgroup intersection identities hold in E_s. Its maximal parabolic orders, in omitted-label order $0, 1, 2, 3$, are $48, 8, 24, 12s^2$.*

Proof. For the fourteen finite supports the certificate checks all 196 exact intersections $E_J \cap E_T = E_{J\cap T}$. For any two elements of these groups with equal linear part, each coordinate of the translation difference has absolute value at most 1. Such a difference belongs to sD only if zero for $s \geq 3$, so these meets survive the quotient.

For every finite support J the certificate also checks $Q_J \cap Q_{012} = Q_{J\cap 012}$, and that each actual affine element of E_J with linear part in Q_{012} satisfies (1). These are exactly the elements of $E_{J\cap 012}$, by injectivity of the finite linear images. Any element of the image of E_J also lying in the image of E_{012} has its linear part in this intersection; it therefore already comes from $E_{J\cap 012}$. Identical supports and the full group give the remaining immediate meets. The orders follow from the table and $[A : sA] = s^2$. □

The toroidal maximal sections are exactly $\{3, 6\}_{(2,0)}$ on (c, b, d) and $\{3, 6\}_{(s,0)}$ on (a, b, d), not just maps with their diagrams. The standard Euclidean reflection group $[3, 6]$ has translation lattice A, generated by the conjugates of $u = a(bd)^2b$; the imposed lattice is exactly sA. This is the usual ordered parameter $(s, 0)$. For the first section the corresponding word with a replaced by c has order two. It is a quotient of $[3, 6]_{(2,0)}$, whose lattice calculation gives order $12 \cdot 2^2 = 48$. Our subgroup has order 48, so the marked epimorphism is an isomorphism. The Euclidean reflection presentation and its torus parameter convention are explicitly retained classical facts. The other maximal sections are C_2^3 and the tetrahedral string group S_4.

6 A faithful action and the literal CPR diagrams

Let

$$w_0 = (1, 1, 1),\ w_1 = (1, -1, -1),\ w_2 = (-1, 1, -1),\ w_3 = (-1, -1, 1).$$

The $4s$ labels (i, k), with $k \in \mathbb{Z}/s\mathbb{Z}$, represent plane families $w_i \cdot x = 2k + 1/2$ modulo period $2s$ in the constant. If $qw_i = \eta w_j$, then $\eta = (-1)^{\epsilon(q)}$ is independent of i. An element (q, v) acts by

$$(i, k) \longmapsto \left(j, \eta k + \frac{\eta - 1 + 2w_j \cdot v}{4}\right). \tag{2}$$

The parity condition makes the fraction integral; translation by sD is trivial. The generators a, b, c transpose family pairs $(1, 2), (2, 3), (0, 3)$ respectively, preserving k. The map d swaps families $1, 2$ with $k \mapsto -k - 1$, and preserves $0, 3$ with $k \mapsto -k$.

Lemma 2. *For $s \geq 3$ this is a faithful action of E_s, with exactly the two parity-dependent CPR diagrams on [1, p. 6].*

Proof. An element fixing all four family labels has $q = I$ or $-I$: these are the kernel of the signed-permutation action on the four diagonal lines. For $q = -I$, fixing both levels $0, 1$ forces $s \mid 2$, impossible. For $q = I$ triviality says $w_i \cdot v \in 2s\mathbb{Z}$ for every i. The vectors $(w_0 + w_i)/2$, $i = 1, 2, 3$, are the coordinate vectors, so $v \in s\mathbb{Z}^3$. The condition on w_0 then gives $v/s \in D$. Thus the kernel in E is exactly sD.

The following explicit labeling was read against the source graphics. Begin with $(0,0)$–$(3,0)$, an edge of label 3. Append $s-1$ alternating four-vertex blocks, each joined to the preceding last vertex by label 1:

$$\begin{aligned} \text{diamond } h: &\quad (2,h),(1,h),(1,-h-1),(2,-h-1),\\ \text{square } h: &\quad (3,-h-1),(0,-h-1),(0,h+1),(3,h+1). \end{aligned}$$

In each diamond label 0 joins first–second and third–fourth, and label 2 joins first–third and second–fourth. In each square label 3 joins first–second and third–fourth; label 2 joins first–fourth and second–third. For $s=2t$, finish with label 1 to $(3,t)$ then label 3 to $(0,t)$. For $s=2t+1$, finish with label 1 to $(2,t)$ then parallel labels $0,2$ to $(1,t)$. All levels are modulo s. The alternating levels $0,-1,1,-2,2,\ldots$ show every label occurs once. The explicit generator formulas give precisely the source's two drawings. This is uniform for both parities; the separately run examples are checks of the transcription, not a proof of the family. IP makes these faithful permutation graphs CPR graphs. □

The range matters: at $s=2$ the nontrivial element $(-I,(1,1,-1))$ fixes all eight labels. No faithful degree-eight claim follows.

7 Flag transitivity

Let $P_i=(E_s)_{I\setminus\{i\}}$. For three subgroups X,Y,Z, Tits' elementary coset criterion says that

$$(X\cap Y)(X\cap Z)=X\cap YZ \tag{3}$$

is equivalent to the three corresponding coset types having the Helly property. First, each P_i has this local property on its three maximal parabolics. The certificate checks (3) on the finite pair subgroups. For supports $7,11,13,14$ the product sizes are $12,12,8,36$ and the intersection size is 4 in each case. The equal-linear-part translation-difference bounds are $1,0,0,0$, so all four identities survive modulo sD for every $s\geq 3$.

For triples containing $P_1=\langle a,d,c\rangle\cong C_2^3$, use

$$(X,Y,Z)=(P_0,P_1,P_2),\ (P_0,P_1,P_3),\ (P_2,P_1,P_3).$$

Indeed $P_1P_2=P_2\cup dP_2$ and $P_1P_3=P_3\cup cP_3$. The extra representative belongs to $X\cap Y$ in each chosen triple, so IP immediately gives (3).

For $(X,Y,Z)=(P_0,P_2,P_3)$ the complete finite S_4 coset decomposition is

$$P_2P_3=P_3\cup cP_3\cup bcP_3\cup abcP_3.$$

The first three representatives lie in $P_0\cap P_2$, so their intersections with P_0 are in the required product. The last coset has empty intersection: P_3 fixes $(0,0)$ in the proved action, whereas abc sends it to $(1,0)$. Its P_0-orbit is exactly $\{(0,0),(3,0),(2,0),(1,-1)\}$, by the three explicit generator formulas. For $s\geq 3$ this excludes $(1,0)$. Hence (3) holds for every triple of maximal parabolics.

For completeness these global and local triple properties imply full four-coset Helly. Normalize one coset to P_0. Its intersections with the other three cosets are cosets in P_0. Any two meet by the global triple property, and the local triple property in P_0 gives a common point of all four. Smaller families are already covered. UA.Geometry therefore gives thinness, residual connectivity and flag transitivity, and identifies E_s with the full type-preserving automorphism group, acting regularly on chambers.

Theorem 1. *For every $s\geq 3$, the literal presentation in [1, Theorem 3.1] defines E/sD of order $48s^3$. Its coset system is the universal locally toroidal regular rank-four hypertope specified there, with exact toroidal parameters $(2,0)$ and $(s,0)$ and the source's faithful degree-4s CPR graph.*

Universality means the exact diagram presentation with the stated local relations: Lemma 1 proves a marked isomorphism, not merely the existence of a finite image.

8 The square family and its affine group

We use column coordinates and composition $(A,t)(B,u) = (AB, t + Au)$. The source writes Euclidean transformations on the right. Reversing an involutory word takes its inverse, so its order and its defining power relation are unaffected by this convention. Let $e_1, \dots, e_4$ be the standard Euclidean basis, and put

$$h = \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4), \qquad \Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}h.$$

All integer vectors below are coordinates in the ordered basis (e_1, e_2, e_3, h) of Λ.

Take the five source reflections, here specified in Euclidean coordinates:

$$\begin{aligned}
R_0(x) &= (1 - x_1, x_2, x_3, x_4),\\
R_1(x) &= x - \tfrac{1}{2}(x_1 - x_2 - x_3 - x_4)(1, -1, -1, -1),\\
R_2(x) &= (x_1, x_2, x_3, -x_4),\\
R_3(x) &= (x_1, x_2, x_4, x_3),\\
R_4(x) &= (x_1, x_3, x_2, x_4).
\end{aligned}$$

Set $E = \langle R_0, \dots, R_4 \rangle$ and mark it by

$$t_0 = R_1, \quad t_1 = R_2, \quad t_2 = R_1R_3, \quad t_3 = R_0R_4. \tag{S.1}$$

These transformations preserve Λ. They generate E: besides $R_1 = t_0$ and $R_2 = t_1$, one has

$$R_3 = t_0t_2, \qquad R_4 = (t_0t_3)^3, \qquad R_0 = t_3(t_0t_3)^3. \tag{S.2}$$

The last two identities follow from the commuting pair R_0, R_4, the commuting pair R_1, R_4 and $(R_1R_0)^3 = 1$.

For $J \subseteq \{0,1,2,3\}$ write $E_J = \langle t_j : j \in J \rangle$. Let Q_J be its linear image and K_J its translation kernel. The full linear image Q has order 1152. A faithful finite action for its calculation is the action on the following 48 Euclidean roots:

$$\{\pm e_i\} \ \cup \ \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\} \ \cup \ \{\pm e_i \pm e_j : i < j\}. \tag{S.3}$$

The roots span the space, so fixing them all forces a linear map to be the identity.

The exact affine computation gives the following complete table. The two nonsingleton translation lattices have bases displayed below it.

J	$\lvert Q_J\rvert$	K_J	J	$\lvert Q_J\rvert$	K_J
$\varnothing$	1	0	$\{3\}$	2	0
$\{0\}$	2	0	$\{0,3\}$	12	0
$\{1\}$	2	0	$\{1,3\}$	4	0
$\{0,1\}$	6	0	$\{0,1,3\}$	48	0
$\{2\}$	2	0	$\{2,3\}$	6	0
$\{0,2\}$	4	0	$\{0,2,3\}$	36	0
$\{1,2\}$	12	0	$\{1,2,3\}$	36	K
$\{0,1,2\}$	48	0	$\{0,1,2,3\}$	1152	$\mathbb{Z}^4$

Here

$$K = \mathbb{Z}u + \mathbb{Z}v, \qquad u = (1,-1,1,0), \quad v = (0,-1,0,1). \tag{S.4}$$

In particular the full translation lattice of E is exactly Λ. The entries are obtained by enumerating each finite Q_J, retaining affine word lifts and computing all normalized Schreier translations, as in AL.1 of the common paper. Integer Hermite reduction computes their entire generated

lattice. It is essential here to use normalized transitions: a list of unnormalized differences alone would not suffice.

The table also exhibits an integral direct complement to K. Writing f_i for the integer coordinate vectors, the four columns u, v, f_2, f_3 form a unimodular matrix. More explicitly, every $x = (x_1, x_2, x_3, x_4) \in \mathbb{Z}^4$ has the unique expression

$$x = x_1 u + x_4 v + (x_2 + x_1 + x_4) f_2 + (x_3 - x_1) f_3. \tag{S.5}$$

Consequently $K \cap s\mathbb{Z}^4 = sK$ for every positive integer s.

9 The presentation and the period

Put $a = t_0, b = t_1, c = t_2, d = t_3$. Let $\mathcal{U}$ be the abstract group on four generators with their involution relations and

$$\begin{gathered}(ab)^3 = (ac)^2 = (ad)^6 = (bc)^6 = (bd)^2 = (cd)^3 = 1, \\ \big(c(da)^2\big)^2 = \big(b(ad)^2 a\big)^2 = \big(a(bc)^2 b\big)^2 = 1.\end{gathered} \tag{S.6}$$

These are the square diagram relations and the three fixed toroidal residue relations. All hold in the affine transformations. The marked map $\mathcal{U} \to Q$ is onto. Its exact Reidemeister–Schreier presentation, followed by Tietze simplification, has four kernel generators and the following six relators:

$$\begin{aligned} &z_4^{-1} z_1^{-1} z_4 z_1, \quad z_2 z_1^{-1} z_2^{-1} z_1, \quad z_4^{-1} z_2 z_4 z_2^{-1}, \\ &z_3 z_1 z_3^{-1} z_1^{-1}, \quad z_3^{-1} z_4^{-1} z_3 z_4, \quad z_3^{-1} z_2 z_3 z_2^{-1}. \end{aligned} \tag{S.7}$$

Thus the actual kernel, not merely its abelianization, is free abelian of rank four. The affine translation kernel has the same rank by the preceding table. AL.2 now identifies $\mathcal{U}$ with E, with the marking preserved.

Define $w = d(cb)^2$. Direct affine multiplication gives

$$w^2 = (I, v). \tag{S.8}$$

The Q-orbit of v consists of 24 vectors and spans $\mathbb{Z}^4$ over the integers. This is checked by its full integral Hermite basis I_4. Conjugating a translation by an affine transformation applies its linear part to the translation vector. Hence the normal closure of w^{2s} in E is exactly $s\Lambda$, and

$$\mathcal{U} / \langle\langle w^{2s} \rangle\rangle \cong E / s\Lambda, \qquad |E / s\Lambda| = 1152 s^4. \tag{S.9}$$

This proves a marked presentation identity for all s at once.

For comparison with the source's original toroid, the five-generator Coxeter presentation $[3, 3, 4, 3]$ has the same affine realization. The separate exact kernel calculation over Q again reduces to four generators with all six pairwise commutator relations and no others. AL.2 identifies that presentation with E. Its period word satisfies

$$\big(R_0 R_1 (R_2 R_3 R_4)^3\big)^2 = (I, f_4).$$

The Q-orbit of f_4 also spans $\mathbb{Z}^4$. Its sth power therefore imposes exactly $s\Lambda$. Together with (S.2), this proves that the groups in Proposition 4.1 are the same marked constructions under the displayed change of generators, not just groups of the same order.

10 The ordered residues

Let P_i be the image of $E_{I\setminus\{i\}}$ in $E/s\Lambda$. For $i = 1, 2, 3$ the linear maps are injective on the entire affine parabolics, so their groups do not collapse in any translation quotient. In the respective ordered paths

$$(t_2, t_3, t_0), \qquad (t_1, t_0, t_3), \qquad (t_0, t_1, t_2),$$

the edge orders are $3, 6$ and the end generators commute. Their additional relations in (S.6) are precisely those of $\{3, 6\}_{(1,1)}, \{3, 6\}_{(2,0)}, \{3, 6\}_{(2,0)}$. The standard triangle presentations with these relations have orders $36, 48, 48$, respectively; exact finite presentation calculations give these orders. The table gives the same orders for the three images, so the marked epimorphisms are isomorphisms. No classification by order alone is being used.

For P_0, put $(A, B, C) = (t_3, t_2, t_1)$ and let

$$\Delta = \langle A, B, C \mid A^2 = B^2 = C^2 = (AB)^3 = (BC)^6 = (AC)^2 = 1\rangle.$$

Its epimorphism onto the actual linear group $Q_{\{1,2,3\}}$ of order 36 has kernel with the exact simplified presentation

$$\langle z_1, z_2 \mid z_2 z_1^{-1} z_2^{-1} z_1 = 1\rangle.$$

The affine image has translation lattice K of rank two. The restriction from this kernel onto K is a surjection of free abelian groups of the same rank, hence an isomorphism. This is the same rank argument as AL.2, now with rank two inside a four-dimensional ambient affine space. Thus $E_{\{1,2,3\}}$ is the marked triangle group Δ itself.

The $Q_{\{1,2,3\}}$-orbit of v has six vectors and spans exactly K. It follows that imposing $(A(BC)^2)^{2s} = 1$ on Δ factors its translation lattice by sK. By (S.5) the ambient quotient induces exactly the same subgroup sK, without any extra residue collapse. This is the standard marked presentation of $\{3, 6\}_{(s,s)}$, and its order is $36s^2$. The ordered rank-three groups are therefore exactly the four claimed in Proposition 4.1.

The source also computes two word orders to distinguish the possible residue periods. Here (S.8) has order s, while $A(BC)^2B$ has linear part of order three and cube (I, u). The primitive vector u has order s modulo $s\Lambda$, so this word has order $3s$. The word itself is generally not a translation: it is its cube that is a translation. This corrects the terminology in the source without changing either asserted order.

11 Intersections and flag transitivity for every period

The fixed linear marking has the intersection property. The certificate checks all 16^2 equalities $Q_J \cap Q_T = Q_{J\cap T}$ as literal sets of matrices. Its four maximal parabolic coset sets have $32, 32, 24, 24$ members. Complete enumeration of pairwise incident cosets gives the following numbers after translating the first coset to the subgroup itself:

types	012	013	023	123	0123
pairwise incident families	18	18	18	18	36.

Every listed family has a common element. Rank-one and rank-two families already satisfy that assertion by nonemptiness and incidence. This proves typed coset Helly for the fixed linear quotient. The normalization loses no family, because right translation acts transitively on the cosets of the first type.

Now fix a positive integer s and set $G = E/s\Lambda$, with the induced map $q : G \to Q$. Its kernel is $N = \Lambda/s\Lambda$. For a proper marked parabolic G_J, the table and the fact that $s\Lambda$ consists of translations give

$$N \cap G_J = \begin{cases} (K + s\Lambda)/s\Lambda, & J = \{1, 2, 3\}, \\ 1, & \text{every other proper } J. \end{cases} \tag{S.10}$$

For the full index set the intersection is N. This entire kernel profile has the intersection property: two different proper supports cannot both have a nontrivial kernel intersection. UA.K therefore proves the intersection property in G.

Equation (S.5) yields the internal direct product

$$N \cong (K/sK) \ \oplus \ \big((\mathbb{Z}f_2 + \mathbb{Z}f_3)/s(\mathbb{Z}f_2 + \mathbb{Z}f_3)\big).$$

Each maximal parabolic intersection in (S.10) is a product of some of these two factors. The linear quotient has IP and typed coset Helly, so UA.SK.Helly transfers typed coset Helly to G. Its four markers have order exactly two, as their nontrivial linear images do. UA.Geometry now proves that the coset incidence system is thin, residually connected and flag-transitive, and that its full type-preserving automorphism group is G.

Finally (S.6) and (S.9) give exactly the diagram relations and the four ordered residue relations. There is no extra global relation. This proves universality, completing Theorem 4.3 for its full stated range $s \geq 2$. The argument in fact also verifies these group and coset assertions at $s = 1$; no additional claim about the terminology for the source's rank-five toroid at that parameter is needed.

12 The literal presentations and a small countermodel

Here the order of the markers is $a = \rho_0, b = \rho_1, c = \rho_2, d = \rho_3$. Let B denote the following thirteen words, all imposed as relators:

$$a^2, b^2, c^2, d^2, (ab)^3, (ac)^6, (ad)^2, (bc)^2, (bd)^6, (cd)^3,$$
$$(d(ca)^2c)^2, \quad (a(bd)^2)^2, \quad (b(ac)^2a)^2.$$

Put $w = c(db)^2d$ and $v = c(db)^2$. For $t \in \{1, 2\}$ define

$$P_t = \langle a, b, c, d \mid B, w^{3t} \rangle, \qquad V_t = \langle a, b, c, d \mid B, v^{2t} \rangle. \tag{F.1}$$

Thus P_1, P_2 are the original presentations. The printed replacement on p. 10 has an extra closing parenthesis, but removes the final d and uses exponent two; p. 11 unambiguously uses v^4. The assertions that these replacements imply $(bd)^2 = 1$ and produce the stated 240-chamber polytope are false for these literal words.

Here is one countermodel for both, in the symmetric group on eleven points; multiplication is functional composition, right factor first:

$$a = (1\,3)(2\,4)(5\,6)(7\,9), \qquad b = (1\,2)(3\,5)(4\,6)(7\,8),$$
$$c = (3\,6)(4\,5)(9\,10), \qquad d = (1\,4)(2\,3)(10\,11).$$

Direct substitution makes every word of B, v^2, and v^4 the identity, while bd has order six. In particular $(bd)^2$ sends 1 to 4. Closing these four permutations under multiplication gives 720 distinct elements. Hence neither V_t can have order 240: each has a quotient of order 720. The complete integer-permutation check is `finite/countermodel.py`. Six declarations in `finite/formal/SectionFive.lean` prove the thirteen shared relations, both replacement relations, the failure of $(bd)^2 = 1$, and both resulting universal nonconsequences. The order 720 computation is not included in the Lean scope.

13 Exact corrected groups and ordered residues

Within P_t, for an omitted index i let G_i be the subgroup generated by the other three markers. We write $T(s, 0)$ for the marked triangular torus group with presentation

$$\langle x, y, z \mid x^2, y^2, z^2, (xy)^3, (yz)^6, (xz)^2, (x(yz)^2y)^s \rangle,$$

and $T(t,t)$ for the same first six relators followed by $(x(yz)^2)^{2t}$. These are the usual triangle-reflection torus conventions in [1]; their orders are $12s^2$ and $36t^2$.

Theorem 2. *For $t = 1, 2$, the presentations P_t and V_t are isomorphic by the map fixing all four named generators, and $|P_t| = 720t^6$. Their ordered maximal parabolics are as follows.*

Omitted index	*Ordered markers (x, y, z)*	*Marked group*	*Order*
0	(c, d, b)	$T(t,t)$	$36t^2$
1	(d, c, a)	$T(2,0)$	*48*
2	(a, b, d)	$T(1,1)$	*36*
3	(b, a, c)	$T(2,0)$	*48*

The marked map $T(3t, 0) \longrightarrow G_0$ is an epimorphism with kernel of order three. The associated torus-map covering has degree three.

Proof. The complete required program `finite/exact_groups.g` constructs each abstract group in (F.1) and calls GAP's exact fp-group isomorphism routine, not just a homomorphism onto an assumed finite image. It obtains orders 720 and 46080. In the original presented group it verifies $v^{2t} = 1$; in the modified presented group it verifies $w^{3t} = 1$. Thus the two identity-on-generators homomorphisms exist and are inverse. Exact GAP fp-group and coset computations are declared computational premises of this proof.

For each row of the table, the program constructs the displayed three-generator abstract torus presentation, checks its order, and constructs a homomorphism to the actual parabolic with exactly the listed generator images. This map is onto and both groups have the listed order, so it is a marked isomorphism, not identification from orders alone. The analogous map from $T(3t, 0)$ is onto G_0; its domain order $108t^2$ gives kernel size three. These are the standard triangular translation lattices, with no local rank-two collapse: each relevant marked dihedral subgroup has its prescribed order. Hence the torus map is a covering of degree three.

For a second, elementary finite check, `finite/model_data.json` records faithful images on 13 and 30 points. `finite/finite_models.py` closes these literal permutations, substitutes every original and modified relator, and verifies all parabolics and all tests below. Its closure gives the finite-image lower bound only; the abstract upper bound still comes from the separate exact fp-group computation above. □

14 Intersection, flags and universality

Lemma 3. *For three subgroups H, K, L of a group, all pairwise intersecting triples of their right cosets have a common point if and only if*

$$H \cap KL = (H \cap K)(H \cap L). \tag{F.2}$$

If four subgroup types satisfy this for every triple, and the three intersection subgroups inside any one type also satisfy it, every pairwise intersecting four-coset family has a common point.

Proof. Translate an intersecting pair of cosets to H, K. Choose $h \in H \cap Lx$ and $k \in K \cap Lx$. Then $hk^{-1} \in L$, so $h \in H \cap LK$. Taking inverses in (F.2) gives $h = l_0k_0$ with $l_0 \in H \cap L$ and $k_0 \in H \cap K$. Now $k_0 = l_0^{-1}h \in Lx$ is a common point. Conversely the common-point property applied to these normalized triples gives the factorization. For four cosets, intersect each of the other three with a fixed fourth. Global triple intersection makes these three cosets inside the fourth pairwise intersecting. The local three-coset property supplies their common point. □

For each of the two actual finite groups, both required checks enumerate all 16 generated subgroups G_J, including $G_\varnothing = 1$, and test all 256 equations

$$G_J \cap G_K = G_{J \cap K}. \tag{F.3}$$

They also test (F.2) for all four triples of maximal parabolics, and for the three subgroups $G_i \cap G_j$ inside each G_i (four more tests). All equalities hold as full sets, not merely as cardinalities. Lemma 3 therefore proves the coset geometry flag-transitive in every rank up to four.

The markers are nontrivial involutions. Equation (F.3) and the full typed coset Helly property therefore satisfy UA.Geometry of the common paper. The coset system is thin, residually connected and flag-transitive, and the displayed group is its full type-preserving automorphism group.

The modified presentation V_t comprises precisely the circuit relations and the four displayed ordered rank-three period relations. By the marked isomorphism, P_t is therefore also the universal marked group with those local presentations. In particular an alternative compatible marked regular geometry is a quotient; the local maps are isomorphisms when its parabolics have those exact presentations. Neither literal replacement changes this group. The distinction between ordered local constraints and an unordered list of toroidal maps is retained throughout; we do not claim to classify every permutation of the local data.

15 The relation with the full cubic toroid

Let Q be the signed permutation group on three coordinates and $D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}$. In the notation of the reviewed cubic-family proof, the hypertope group is $E_s = E/sD$, where

$$E = \{(q, v) \in Q \ltimes \mathbb{Z}^3 : \textstyle\sum v_i \equiv \epsilon(q) \pmod 2\}, \qquad |E_s| = 48s^3.$$

The introductory sentence on page5 of [1] identifies this group with the full group of the cubic toroid $\{4, 3, 4\}_{(s,s,0)}$. The earlier account [2, pp. 14–15] correctly describes an index-two subgroup. Here is the exact change of generators and coordinates.

The standard Euclidean realization of $[4, 3, 4]$ is generated by

$$R_0(x) = (1 - x_1, x_2, x_3), \qquad R_1(x) = (x_2, x_1, x_3),$$
$$R_2(x) = (x_1, x_3, x_2), \qquad R_3(x) = (x_1, x_2, -x_3).$$

Its affine image is $W = Q \ltimes \mathbb{Z}^3$: the last three generators give Q, and multiplication of R_0 by the corresponding linear reflection gives translation by e_1. Conjugating gives all coordinate translations. The faithful Euclidean Coxeter realization is a retained classical fact. Direct multiplication gives

$$(R_0R_1R_2R_3R_2)^2 = (I, e_1 + e_2).$$

Its Q-orbit generates D. Thus the additional relation with exponent $2s$ used in [2] imposes precisely sD, and the full toroid group is W/sD, of order $48[\mathbb{Z}^3 : sD] = 96s^3$.

Put $p = (1/2, 1/2, 1/2)$ and $T(x) = \mathrm{diag}(1, -1, -1)(x - p)$. Conjugating the four displayed ambient words by T gives

$$T(R_0R_1R_0)T^{-1}(x) = (x_2, x_1, x_3),$$
$$TR_2T^{-1}(x) = (x_1, x_3, x_2),$$
$$T(R_1R_3)T^{-1}(x) = (-x_2, -x_1, 1 - x_3),$$
$$TR_1T^{-1}(x) = (-x_2, -x_1, x_3).$$

These are exactly the four generators of E in source-label order. The linear part of T preserves D, so this conjugacy descends to the period quotients. Consequently the displayed subgroup of W/sD is isomorphic to E/sD and has index two. In [2] the Petrie generators are $\alpha_0 = R_0$, $\alpha_1 = R_1$, $\alpha_2 = R_2$ and $\alpha_3 = R_1R_3$; its four subgroup generators are the same set, ordered here as $(\alpha_1^{\alpha_0}, \alpha_2, \alpha_3, \alpha_1)$ to match [1].

The family still arises from the stated toroid. The correction is that its group is the displayed index-two subgroup, rather than the full toroid group. This changes no principal result of Section3 of [1]. It does not claim a new family or a resolution of an open question.

Whole dependencies and formal limits. Certificate paths in the transferred sections refer to the entire corresponding files under `inherited/`; their exact source bytes are preserved. The new finite-row engine is separately listed in the application. GAP's exact finite-presentation, coset and permutation algorithms and the bounded ACE enumeration are declared software premises. The inherited six Lean declarations concern only the finite countermodel and nonconsequences. They are not a formal proof of the general affine or C-plus geometry theorems. No numerical or source-credit assertion is made by this relocation.

References

[1] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, retained twelve-page author edition.

[2] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition.

[3] *Supported kernels, translation lattices and finite marked quotients*, complete accepted common manuscript, AL.1–AL.2, AA.ParabolicPeriods, UA.K, UA.SK.Helly and UA.Geometry.

The seven shared group structures in the hexagonal tables

A structural supplement to the accepted marked affine models

We recover the group-column assertions in Table 6, rows 2–5, and Table 8, rows 1–3, of Fernandes–Leemans–Ivić Weiss [1]. The original constructions and table assertions remain theirs. The complete earlier marked-kernel common theory and the paid *Rotation certificates and affine period families* supplement remain premises. In particular, the universal presentations, marked residues, intersection property and flag geometry are already proved there. Here we use those exact models to supply previously missing normal factors and splittings. We do not rerun their presentation or geometry computations.

The source adopts Atlas notation in its introduction. A colon records a split extension and a cross a direct product. Iterated extensions must be interpreted at the indicated stages: a complement at each stage does not imply a complement to the innermost kernel in the entire group. The actions in the source abbreviations are unnamed. Our realizations specify them by the accepted faithful affine actions and, for the fixed finite cases, explicit permutation generators and bijective maps. An unspecified action never replaces a proof of a kernel, quotient or splitting assertion.

1 One splitting for every cubic parameter

Let Q be the signed permutation group on three coordinates and let $\epsilon(q) \in \{0,1\}$ be the parity of its number of sign changes. The accepted cubic model is

$$D = \{v \in \mathbb{Z}^3 : \textstyle\sum v_i \equiv 0 \pmod 2\}, \qquad E = \{(q,v) \in Q \ltimes \mathbb{Z}^3 : \textstyle\sum v_i \equiv \epsilon(q) \pmod 2\}.$$

Products act on column vectors, with $(q,v)(r,w) = (qr, v + qw)$. The literal full group at parameter $s \geq 3$ is $G_s = E/sD$, of order $48s^3$. Its marking is the exact marking in the shared proof; no new isomorphism based merely on this order is used.

Lemma 1. *For every positive integer s, the abstract affine quotient has the split form*

$$E/sD \cong (D/sD) \rtimes Q.$$

For odd s this is $\mathrm{Dih}(2s) \wr S_3$*, in the imprimitive wreath action with three factors, where* $\mathrm{Dih}(2s)$ *has order* $2s$*.*

Proof. Put $p = (1/2, 1/2, 1/2)$. For every signed permutation q, the vector $p - qp$ is integral: its entries are zero or one, with a one exactly where the corresponding sign is negative. Thus $\sum_i (p - qp)_i \equiv \epsilon(q) \pmod 2$, and

$$\sigma(q) = (q, p - qp) \in E, \qquad \sigma(q)\sigma(r) = \sigma(qr).$$

The group $\sigma(Q)$ fixes p and meets the translation subgroup D trivially. An arbitrary $(q,v) \in E$ differs from $\sigma(q)$ by a translation whose coordinate sum is even. Hence $E = D \rtimes \sigma(Q)$. Quotienting by the invariant subgroup sD preserves this complement, proving the first assertion for every s.

If s is odd, reduction of coordinates gives a Q-equivariant isomorphism $D/sD \longrightarrow (\mathbb{Z}/s\mathbb{Z})^3$. Indeed, any coordinate residue has an even-sum lift after adding s to one coordinate if necessary. If $sw \in D$, oddness of s implies $w \in D$, proving injectivity. The three independent sign changes invert the three cyclic factors, and S_3 permutes them. This is precisely the displayed wreath product, including its action. □

It follows at once that Table 6 row 2 ($s = 3$) has

$$G_3 \cong (S_3 \times S_3 \times S_3) \rtimes S_3,$$

and row 4 ($s = 5$) has

$$G_5 \cong C_5^3 \rtimes (C_2^3 \rtimes S_3).$$

The three commuting sign factors refine the latter to the source's $5^3 : 2 : 2 : 2 : S_3$. For $s = 6$, the Chinese remainder isomorphism of the free lattice is Q-equivariant:

$$D/6D \cong D/2D \oplus D/3D.$$

The subgroup corresponding to the second summand together with the same $\sigma(Q)$ is a complement to $D/2D$. Consequently row 5 has

$$G_6 \cong C_2^3 \rtimes \big((S_3 \times S_3 \times S_3) \rtimes S_3\big).$$

This gives the successive split factors in the source's $2^3 : S_3 : S_3 : S_3 : S_3$. It makes no assertion that the action on the first C_2^3 is trivial.

2 The even cubic row and the fixed square group

For the remaining cubic parameter $s = 4$, write $T = D/4D$ and let $X \cong C_2^3$ be the three sign changes inside $\sigma(Q)$. Use the accepted faithful action on the $4s = 16$ plane labels. In that action set

$$N = \langle T^2,\ t_{(2,0,0)},\ X \rangle, \qquad C = \langle a, b, c \rangle,$$

where the accepted ordered marking is (a, b, d, c). These are explicit subgroups of the already identified full group G_4. The complete small certificate gives

$$|N| = 128, \quad Z(N) = N' = \Phi(N) \cong C_2, \quad N \trianglelefteq G_4, \qquad C \cong S_4, \quad N \cap C = 1, \quad NC = G_4.$$

Thus N is extraspecial, by its defining centre, derived and Frattini conditions. Inside the actual complement C, the normal Klein four group and a point stabilizer give $C = C_2^2 \rtimes S_3$. Table 6 row 3 therefore has the asserted structure

$$G_4 \cong 2^{1+6} : (2^2 : S_3).$$

The normality and complement identities, and the bijections to S_4 and S_3, are checked on these actual subgroups. They are not inferred from the orders 128, 24 and 6.

For the square family the accepted lattice and generators are

$$\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}h, \qquad h = \tfrac{1}{2}(e_1 + e_2 + e_3 + e_4),$$

$$R_0(x) = (1 - x_1, x_2, x_3, x_4), \qquad R_1(x) = x - \tfrac{1}{2}(x_1 - x_2 - x_3 - x_4)(1, -1, -1, -1),$$

with R_2 negating x_4, R_3 interchanging x_3, x_4, and R_4 interchanging x_2, x_3. The earlier proof identifies $E_4 = \langle R_0, \ldots, R_4 \rangle$, its entire translation kernel Λ, and the source's full marked group as $E_4/s\Lambda$.

Theorem 2. *For every $s \geq 1$, Table 8 row 1 has the split structure*

$$E_4/s\Lambda \cong C_s^4 \rtimes \big(2^{1+4} \rtimes (S_3 \times S_3)\big).$$

It therefore realizes the successive source factors $s^4 : 2^{1+4} : S_3 : S_3$ uniformly.

Proof. The subgroup $Q_0 = \langle R_1, R_2, R_3, R_4 \rangle$ fixes the origin. Its action on the following 48 roots is faithful, since they span the ambient vector space:

$$\{\pm e_i\} \ \cup \ \{\tfrac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\} \ \cup \ \{\pm e_i \pm e_j : i < j\}.$$

The fixed certificate constructs these four permutations from the literal displayed linear maps. It verifies $|Q_0| = 1152$, equal to the entire linear image already proved for E_4. Thus Q_0 maps onto that image and meets the translation kernel trivially; consequently $E_4 = \Lambda \rtimes Q_0$. Its complement still injects after quotienting by every $s\Lambda$, including $s = 1$.

In this one fixed faithful Q_0, the certificate finds its normal 2-core N_0 and an actual complement C_0, and verifies

$$|N_0| = 32, \quad Z(N_0) = N_0' = \Phi(N_0) \cong C_2, \quad Q_0 = N_0 C_0, \quad N_0 \cap C_0 = 1.$$

An explicit bijective homomorphism identifies C_0 with $S_3 \times S_3$; the inverse images of the two factors commute, intersect trivially and generate C_0. Therefore N_0 is extraspecial and the fixed linear structure is proved. Combining it with the uniform affine splitting proves the theorem. No list of sampled values of s is used. □

3 The two inherited finite presentations

Let H_1, H_2 be the exact full marked groups of orders 720 and 46080 in the accepted L121 finite proof. Its complete 13- and 30-point models are copied byte for byte as `inputs/model_data.json`. Their universal orders and geometry remain the accepted premises. The new program uses only these faithful models for structural witnesses.

For H_1 it constructs the normal perfect residual A and centre Z, identifies $A \cong A_5$ bijectively, and verifies

$$|Z| = 2, \quad A \cap Z = 1, \quad AZ = A \times Z \trianglelefteq H_1.$$

It then gives a subgroup $C \cong S_3$ with $C \cap AZ = 1$ and $(AZ)C = H_1$. This proves the printed Table 8 row 2 structure $(A_5 \times 2) : S_3$.

For H_2, let P be its normal perfect residual and Z its centre. The complete witnesses are

$$|P| = 960, \quad |Z| = 2, \quad P \cap Z = 1, \quad P = V \rtimes A, \quad V \cong C_2^4, \quad A \cong A_5,$$
$$PZ = P \times Z \trianglelefteq H_2, \qquad H_2 = (P \times Z) \rtimes C, \qquad C \cong S_4.$$

The proof checks elementary abelianness of the actual V, normality, all complement intersections and products, and both named bijections. To make the source's mixed string precise, put $C = W \rtimes S$ with $W \cong C_2^2$ and $S \cong S_3$, using the usual Klein four and point stabilizer through the verified S_4 map. Choose two independent involutions u, v in W. The program verifies the successive identities

$$K_1 = P \rtimes \langle u \rangle, \qquad K_2 = K_1 \times Z, \qquad K_3 = K_2 \rtimes \langle v \rangle, \qquad H_2 = K_3 \rtimes S.$$

Their orders are $1920, 3840, 7680, 46080$, respectively. Together with $P = 2^4 : A_5$, this realizes Table 8 row 3 as $2^4 : A_5 : 2 \times 2 : 2 : S_3$, with every operation given its indicated stage. We do not assert that the whole group splits over V; that stronger assertion is not needed for this iterated structure.

4 The exact finite premise and its limits

The complete new mathematical programs are `prepare_models.py` and `verify.g`; `run.py` provides their portable replay. The first constructs the faithful cubic plane actions and square root action directly from the accepted coordinates and copies the accepted finite models. The second records all subgroup generators, named isomorphism images, quotient images, and complement data. Each split extension checks normality, trivial intersection and generation, and verifies that the complement maps bijectively onto the actual quotient. Each direct product checks commutation as well. Extraspecial notation is justified by equality of the centre, derived and Frattini subgroups of order two. GAP's exact finite permutation, normal-subgroup, complement and isomorphism algorithms are retained computational trust. A structure-description string is never a proof premise.

The checks involve the four cubic images, one fixed linear group, and two inherited finite groups only. They do not repeat the 66-row presentation census, the coset-geometry checks or the old Lean work. The uniform cubic and square assertions have the written lattice and splitting proofs above; the finite certificates establish only their fixed structural components. All seven rows preserve their accepted markings and geometry. No other printed group column, source question, whole-original admission, numerical factor or new formalization follows from this supplement.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *Hexagonal extensions of toroidal maps and hypermaps*, retained 21-page author edition, Tables 6 and 8, physical pp. 15 and 18.

[2] *Rotation certificates and affine period families*, complete accepted marked-model supplement, sections 4–15; the earlier *Supported kernels, translation lattices and finite marked quotients* remains fully acquired.

[3] M. E. Fernandes, D. Leemans, C. A. Piedade and A. Ivić Weiss, *Two families of locally toroidal regular 4-hypertopes arising from toroids*, retained twelve-page author edition; exact corrected proofs and finite models as separately accepted and migrated.